

Online Resource 1: Supplementary Material for “Semiparametric Inference and Sensitivity Analysis for Causal Mediation in Linear Models with Unspecified Error Distributions”

Mijeong Kim

Department of Statistics, Ewha Womans University, Seoul, Republic of Korea

m.kim@ewha.ac.kr

ORCID: 0000-0002-3578-5413

Prepared for the Journal of the Korean Statistical Society

This supplement follows the main article’s order, supplying theoretical derivations, implementation details, complete simulation results and JOBS II diagnostics. The main article gives the principal interpretation; the tables here retain effect-specific estimates, calibration and numerical-reliability measures. Section S10 describes reproduction of the analyses.

S1 Identification calculation

Under sequential ignorability, consistency, and positivity, the mediation formula identifies the mean nested potential outcomes. The linear models imply

$$\mathbb{E}\{M(t) \mid \mathbf{X}\} = \alpha_2 + \beta_2 t + \boldsymbol{\xi}_2^\top \mathbf{X}$$

and

$$\mathbb{E}\{Y(t, m) \mid \mathbf{X}\} = \alpha_3 + \beta_3 t + \gamma m + \eta t m + \boldsymbol{\xi}_3^\top \mathbf{X}.$$

Therefore, conditional on $\mathbf{X}$,

$$\mathbb{E}[Y\{t, M(1)\} - Y\{t, M(0)\} \mid \mathbf{X}] = (\gamma + \eta t)\beta_2,$$

$$\mathbb{E}[Y\{1, M(t)\} - Y\{0, M(t)\} \mid \mathbf{X}] = \beta_3 + \eta\{\alpha_2 + \beta_2 t + \boldsymbol{\xi}_2^\top \mathbf{X}\}.$$

Averaging over $\mathbf{X}$ yields

$$\text{PNIE} = \beta_2 \gamma,$$

$$\text{TNIE} = \beta_2(\gamma + \eta),$$

$$\text{PNDE} = \beta_3 + \eta\mu_0,$$

$$\text{TNDE} = \beta_3 + \eta\mu_1,$$

where $\mu_0 = \alpha_2 + \boldsymbol{\xi}_2^\top \boldsymbol{\mu}_X$ and $\mu_1 = \mu_0 + \beta_2$. Direct addition gives

$$\text{PNDE} + \text{TNIE} = \text{TNDE} + \text{PNIE} = \beta_3 + \beta_2 \gamma + \eta\mu_1 = \text{TE}.$$

S2 Gradient of the natural-effect map

Write the parameter blocks as

$$\theta_M = (\alpha_2, \beta_2, \boldsymbol{\xi}_2^\top, v_M)^\top, \quad \theta_Y = (\alpha_3, \beta_3, \gamma, \eta, \boldsymbol{\xi}_3^\top, v_Y)^\top,$$

where the implementation uses log variances without changing any effect derivative. Gradients not listed below are zero. With differentials used to display the full map,

$$\begin{aligned}
d\text{PNIE} &= \gamma d\beta_2 + \beta_2 d\gamma, \\
d\text{TNIE} &= (\gamma + \eta) d\beta_2 + \beta_2 d\gamma + \beta_2 d\eta, \\
d\text{PNDE} &= \eta d\alpha_2 + \eta \boldsymbol{\mu}_X^\top d\boldsymbol{\xi}_2 + \eta \boldsymbol{\xi}_2^\top d\boldsymbol{\mu}_X + d\beta_3 + \mu_0 d\eta, \\
d\text{TNDE} &= \eta d\alpha_2 + \eta d\beta_2 + \eta \boldsymbol{\mu}_X^\top d\boldsymbol{\xi}_2 + \eta \boldsymbol{\xi}_2^\top d\boldsymbol{\mu}_X + d\beta_3 + \mu_1 d\eta, \\
d\text{TE} &= \eta d\alpha_2 + (\gamma + \eta) d\beta_2 + \eta \boldsymbol{\mu}_X^\top d\boldsymbol{\xi}_2 + \eta \boldsymbol{\xi}_2^\top d\boldsymbol{\mu}_X + d\beta_3 + \beta_2 d\gamma + \mu_1 d\eta.
\end{aligned} \tag{S1}$$

Equation (S1) is the analytic matrix G used to propagate the stacked covariance. In particular, the terms involving $d\boldsymbol{\mu}_X$ are retained rather than treating the sample covariate mean as fixed.

S3 Regularity conditions and influence-function composition

Let $W = (\mathbf{X}, T)$. Under the statistical assumptions in the main article, the observed-data density has the factorization

$$p(o) = p_W(w) f_M\{m - m_M(w; \theta_M)\} f_Y\{y - m_Y(w, m; \theta_Y)\}. \tag{S2}$$

The nuisance components are the unrestricted law of W and the two error densities subject to their moment restrictions. Let S_M^{eff} and S_Y^{eff} be the efficient regression scores stated in the main text.

Regularity conditions. For $k \in \{M, Y\}$, let θ_{k0} denote the true regression parameter, including the log variance, and let $\widehat{S}_{k,i}$ denote its feasible score with the density, moments and design expectations replaced by their estimates. Under the regression model and factorization (S2), we use the following conditions. Condition (R1) concerns the population score representation; (R2)–(R4) support asymptotic linearity of the fitted estimators; and (R5) supports covariance estimation and Wald inference.

(R1) **Sampling, regression scores and moments.** The observations are independent and identically distributed, and $\mathbb{E}\|\mathbf{X}\|^2 < \infty$. For each regression, the error density is positive and continuously differentiable on its support, the error has mean zero, positive finite variance and a finite fourth moment, and $0 < Q_{2,k} < \infty$. The location Fisher information is finite, the design has full-rank information, and the population score equation has a locally unique root. The boundary and integrability requirements for the mean-zero score identities and the regression-score calculation hold. In particular, $\mathbb{E}(S_k^{\text{eff}}) = 0$ and $I_k = \mathbb{E}\{S_k^{\text{eff}} S_k^{\text{eff}^\top}\}$ is finite and positive definite. These are the population regression-score requirements stated in Section 2.2 of the main article, together with the moment condition for population averaging over $\mathbf{X}$.

(R2) **Consistent selected roots.** With probability tending to one, each regression has a selected root $\widehat{\theta}_k$ satisfying $\widehat{\theta}_k \xrightarrow{P} \theta_{k0}$, where θ_{k0} is an interior parameter point. The sample mean score at the selected root is $o_p(n^{-1/2})$. An exact root satisfies the latter requirement automatically; for a numerical root it is a condition on the asymptotic solver accuracy.

(R3) **Negligible score-replacement error.** At the true parameter,

$$n^{-1/2} \sum_i \{\widehat{S}_{k,i}(\theta_{k0}) - S_{k,i}^{\text{eff}}(\theta_{k0})\} = o_p(1). \tag{S3}$$

This condition covers the combined replacement of the error density, its derivative, residual moments and design expectations. It controls their effect on the aggregate estimating equation, beyond pointwise consistency of a density or density-derivative estimator. The Gaussian-kernel and leave-one-out construction used in the code is connected to this condition in Proposition S1 below.

(R4) **Local differentiability and derivative convergence.** The sample mean score is continuously differentiable on a neighborhood containing the line segment between θ_{k0} and $\hat{\theta}_k$, and

$$\frac{\partial}{\partial \theta_k^\top} \left\{ n^{-1} \sum_i \hat{S}_{k,i}(\theta_k) \right\} = -I_k + o_p(1) \quad (\text{S4})$$

uniformly on that segment.

(R5) **Covariance consistency and nondegenerate scalar inference.** At the fitted parameters, the empirical stacked score second moments satisfy $\hat{B} \xrightarrow{p} B = \mathbb{E}(\Psi_i \Psi_i^\top)$, and the implemented bread, including its numerical regression derivatives, satisfies $\hat{A} \xrightarrow{p} A = \text{diag}(-I_M, -I_Y, -I_{d_X})$, where d_X is the dimension of $\mathbf{X}$. The effect map g is continuously differentiable. For each scalar effect reported with a regular Wald interval, its limiting variance $G_j A^{-1} B A^{-\top} G_j^\top$ is strictly positive, where G_j is the corresponding row of the effect gradient at the truth. The full effect covariance need not be nonsingular.

Feasible kernel scores and deterministic root selection. To connect the implemented root to Conditions (R2)–(R4), let $\bar{S}_{k,n}^{\text{full}}(\theta)$ be the feasible full-kernel equation of Kim (2023, Theorem 2) and $\bar{S}_{k,n}^{\text{loo}}(\theta)$ its stabilized leave-one-out counterpart. The following proposition gives conditions for first-order equivalence and consistent root selection.

Proposition S1 (Feasible-root bridge). *For each $k \in \{M, Y\}$, suppose the kernel smoothness, boundary, tail, moment and unique-root conditions of Kim (2023, Theorem 2 and its proof) hold uniformly in a neighborhood of θ_{k0} , $s_{e,k} \xrightarrow{p} \sqrt{v_k}$, and $h_{k,n} = 1.06 s_{e,k} n^{-1/5}$. Suppose also that there are expanding central sets $\mathcal{E}_{k,n}$ and constants $c_{k,n} > 0$ such that, uniformly in that neighborhood, the true and estimated densities on $\mathcal{E}_{k,n}$ are at least $c_{k,n}$, the estimated first density derivative is $O_p(1)$, $\kappa_n = \min(10^{-10}, n^{-2}) = o(c_{k,n})$, and*

$$\frac{\sqrt{n}}{n h_{k,n} c_{k,n}^2} \rightarrow 0. \quad (\text{S5})$$

Assume that the contributions from $\mathcal{E}_{k,n}^c$ to the sample-mean score and its derivative are $o_p(n^{-1/2})$ and $o_p(1)$, respectively. Finally, suppose the solver initialized at OLS returns the locally unique valid root with probability tending to one and any other valid root is asymptotically separated from θ_{k0} in the OLS-design distance. Then the selected stabilized leave-one-out root satisfies Conditions (R2)–(R4) and

$$\sqrt{n}(\hat{\theta}_k - \theta_{k0}) = I_k^{-1} n^{-1/2} \sum_{i=1}^n S_{k,i}^{\text{eff}}(\theta_{k0}) + o_p(1). \quad (\text{S6})$$

Consequently, the natural-effect expansion and the OLS-relative variance and local-power comparisons in the main article apply to the feasible estimator returned by the supplied algorithm.

Proof. Write K for the standard Gaussian density and let $\hat{f}_i$ and $\hat{f}'_i$ be the full-sample kernel estimators evaluated at residual e_i . Because $K'(0) = 0$, direct algebra gives

$$\hat{f}_{-i} = \frac{n}{n-1} \left\{ \hat{f}_i - \frac{K(0)}{n h_{k,n}} \right\}, \quad \hat{f}'_{-i} = \frac{n}{n-1} \hat{f}'_i,$$

and hence

$$\frac{\hat{f}'_{-i}}{\hat{f}_{-i}} = \frac{\hat{f}'_i}{\hat{f}_i - K(0)/(n h_{k,n})}.$$

On $\mathcal{E}_{k,n}$, deleting the self-density term therefore changes an observation-level density score by

$$O_p \left\{ (n h_{k,n} c_{k,n}^2)^{-1} \right\}.$$

Equation (S5) makes its root- n aggregate contribution $o_p(1)$. The relation $\kappa_n = o(c_{k,n})$ makes the numerical floor asymptotically inactive on these central sets, and the stated tail condition controls their

complements. The same calculation uniformly in a local neighborhood, together with the derivative condition from the kernel-root argument of Kim (2023, Theorem 2), gives

$$\begin{aligned}\sqrt{n}\{\bar{S}_{k,n}^{\text{loo}}(\theta_{k0}) - \bar{S}_{k,n}^{\text{full}}(\theta_{k0})\} &= o_p(1), \\ \sup_{\theta} \left\| \frac{\partial \bar{S}_{k,n}^{\text{loo}}(\theta)}{\partial \theta^\top} - \frac{\partial \bar{S}_{k,n}^{\text{full}}(\theta)}{\partial \theta^\top} \right\| &= o_p(1),\end{aligned}$$

where the supremum is over the shrinking neighborhood traversed by the root expansion. Thus the leave-one-out root has the same first-order expansion as the full-kernel root in Kim (2023, Theorem 2).

OLS is consistent under Condition (R1), so the distance between the OLS pilot and the local consistent root is $o_p(1)$. By the assumed separation, every valid nonlocal root has distance bounded away from zero with probability tending to one. The minimum-distance rule therefore selects the local root. A mean-value expansion of its score equation yields (S6), which is precisely Conditions (R2)–(R4). Stacking the two expansions with the sample-mean expansion and applying the effect gradient gives the final claim. $\square$

Population influence-function composition. Under Condition (R1) and (S2), every component of the mediator score satisfies

$$\mathbb{E}\{S_M^{\text{eff}} \mid W\} = 0,$$

because both the error-density score and the location–scale correction have zero expectation under f_M . Similarly,

$$\mathbb{E}\{S_Y^{\text{eff}} \mid W, M\} = 0.$$

It follows by iterated expectation that

$$\begin{aligned}\mathbb{E}\{S_M^{\text{eff}} S_Y^{\text{eff}^\top}\} &= 0, \\ \mathbb{E}[S_M^{\text{eff}} \{\mathbf{X} - \boldsymbol{\mu}_X\}^\top] &= 0, \\ \mathbb{E}[S_Y^{\text{eff}} \{\mathbf{X} - \boldsymbol{\mu}_X\}^\top] &= 0.\end{aligned}$$

Thus the two regression scores and the centered baseline covariate are mutually orthogonal in the tangent space generated by (S2). The efficient influence functions for θ_M , θ_Y , and $\boldsymbol{\mu}_X$ are respectively

$$\varphi_M = I_M^{-1} S_M^{\text{eff}}, \quad \varphi_Y = I_Y^{-1} S_Y^{\text{eff}}, \quad \varphi_X = \mathbf{X} - \boldsymbol{\mu}_X.$$

The natural-effect vector is a continuously differentiable finite-dimensional functional,

$$g = g(\theta_M, \theta_Y, \boldsymbol{\mu}_X).$$

Applying the pathwise chain rule in any regular parametric submodel gives

$$\varphi_g = G_M \varphi_M + G_Y \varphi_Y + G_X \varphi_X.$$

This function lies in the tangent space of the model and is orthogonal to its nuisance tangent space while representing the pathwise derivative of g ; hence it is the canonical gradient. Its variance is the population efficiency bound. The result specializes established regression efficiency and the pathwise chain rule to the linked mediation effect map (Kim, 2023; Bickel et al., 1993; Tsiatis, 2006).

OLS-relative variance and local power. Here we give the score-block calculation underlying Proposition 2 in the main article. In one of the regressions, write $R = a + \mathbf{b}^\top \mathbf{W} + \epsilon$, with $\omega = \mathbb{E}(\mathbf{W})$, $C = \text{Var}(\mathbf{W})$, $s(\epsilon) = -f'(\epsilon)/f(\epsilon)$, and $J = \mathbb{E}\{s(\epsilon)^2\}$. Set $q = \epsilon^2 - v - (\mu_3/v)\epsilon$, $Q_2 = \mathbb{E}(q^2)$, and $t = \epsilon/v - \mu_3 q/(vQ_2)$. The efficient score for the intercept is t , the slope score is $s(\epsilon)(\mathbf{W} - \omega) + t\omega$, and the log-variance score is $S_\lambda = vq/Q_2$. Centering the slope parameter by the fixed design mean ω separates its score $s(\epsilon)(\mathbf{W} - \omega)$ from the intercept–variance block. By independence, this slope block has information JC and is orthogonal to the other two scores.

Since $\mathbb{E}(\epsilon q) = 0$, direct calculation gives

$$\mathbb{E}(t^2) = \frac{1}{v} + \frac{\mu_3^2}{v^2 Q_2}, \quad \mathbb{E}(t S_\lambda) = -\frac{\mu_3}{Q_2}, \quad \mathbb{E}(S_\lambda^2) = \frac{v^2}{Q_2}.$$

Residualizing t against S_λ therefore yields

$$t - \frac{\mathbb{E}(t S_\lambda)}{\mathbb{E}(S_\lambda^2)} S_\lambda = t + \frac{\mu_3}{v^2} S_\lambda = \frac{\epsilon}{v},$$

whose information is $1/v$. Joint information inversion thus gives the efficient slope influence function

$$\varphi_b^{\text{semi}} = C^{-1}(\mathbf{W} - \omega) \frac{s(\epsilon)}{J},$$

and the original intercept influence function

$$\varphi_a^{\text{semi}} = \epsilon - \omega^\top \varphi_b^{\text{semi}}.$$

The OLS normal equations give the same two expressions with $s(\epsilon)/J$ replaced by ϵ . Equivalently, with $e_0 = (1, 0, \dots, 0)^\top$ and $H = \begin{pmatrix} -\omega^\top \\ I \end{pmatrix}$,

$$\begin{aligned} \text{Var}(\varphi_\beta^{\text{semi}}) &= v e_0 e_0^\top + J^{-1} H C^{-1} H^\top, \\ \text{Var}(\varphi_\beta^{\text{OLS}}) &= v e_0 e_0^\top + v H C^{-1} H^\top. \end{aligned}$$

The score identity $\mathbb{E}\{\epsilon s(\epsilon)\} = -\int e f'(e) de = 1$ holds under the boundary condition in (R1). Hence $1 \leq \{vJ\}^{1/2}$ and $v - J^{-1} \geq 0$; equality requires $s(\epsilon) = \epsilon/v$ almost surely, the Gaussian density under the stated smoothness and support conditions.

For either estimator, the influence function of natural effect g_j is the sum of the mediator-coefficient, outcome-coefficient, and baseline-mean contributions. The conditional mean-zero properties established above make these three blocks orthogonal for both OLS and the efficient regression estimator. The baseline-mean contribution is identical for the two methods, so subtracting their effect variances yields the OLS-relative variance formula in Proposition 2 of the main article. At a regular local alternative $g_{j,n} = h/\sqrt{n}$, assume the asymptotic linear expansions and variance consistency hold along the contiguous sequence. Studentization gives a limiting $N(h/\sqrt{V_{j,r}}, 1)$ statistic. With $z = z_{1-\alpha_{\text{test}}/2}$ and $\lambda = |h|/\sqrt{V_{j,r}}$, its two-sided rejection probability is $1 - \Phi(z - \lambda) + \Phi(-z - \lambda)$, whose derivative in $\lambda > 0$ equals $\phi(z - \lambda) - \phi(z + \lambda) > 0$. This proves the local-power ordering at a common asymptotic size. The argument does not cover the joint-zero PNIE point, where the first-order gradient vanishes.

Asymptotic inference for the fitted estimators. Under Conditions (R2)–(R4), score expansion using (S3)–(S4) yields (S6). Combining the two regression expansions with $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}_X) = n^{-1/2} \sum_i (\mathbf{X}_i - \boldsymbol{\mu}_X)$ and applying the multivariate delta method proves the estimator part of Proposition 1 in the main article. Condition (R5) gives sandwich consistency and marginal Wald inference for each scalar effect with positive asymptotic variance, while permitting a singular joint effect covariance.

Scope of the conditions and numerical implementation. Proposition S1 requires the OLS-started solver to reach the local valid root with probability tending to one, with other valid roots separated in the selection distance. Local uniqueness, nonsingular score derivatives and a consistent OLS start provide the local setting; finite-sample diagnostics record convergence, conditioning and root multiplicity. The acceptance threshold $b_n = \min(10^{-6}, n^{-1})$ gives mean-score error $o_p(n^{-1/2})$, and the finite-difference factor $d_n = \min(10^{-4}, n^{-1/3})$ vanishes. Together with the derivative and moment conditions, these safeguards support Condition (R5).

S4 Complete numerical algorithm

For each mediator or outcome regression, the supplied function `fit_semiparametric_lm()` performs the following deterministic steps.

1. Fit OLS and compute $(\tilde{\beta}, \tilde{v})$ using $n^{-1} \sum_i \tilde{e}_i^2$.
2. Form coefficient increments $d_j = \max\{0.05, c \max(1, |\tilde{\beta}_j|)\}$ and the seven starts described in the multi-start subsection of the main article.
3. At every solver evaluation, recompute residuals, the leave-one-out density and derivative, the third and fourth residual moments, $q(e_i)$, and all observation-level efficient scores.
4. Run `rootSolve::multiroot()` with relative and absolute tolerances $a_n = \min(10^{-9}, n^{-1})$ and at most 200 iterations.
5. Retain a root if it is finite, its maximum absolute mean score is at most $b_n = \min(10^{-6}, n^{-1})$, its centered finite-difference Jacobian (coordinate step $d_n \max\{1, |\theta_j|\}$, $d_n = \min(10^{-4}, n^{-1/3})$) has reciprocal condition number at least 10^{-10} , and its sandwich covariance is finite with positive diagonal.
6. Remove duplicate roots after rounding parameter vectors to seven decimal places.
7. Select the valid root with minimum OLS-design distance defined in the main article. When the candidate set is empty, record a numerical failure so every reported proposed-method estimate remains a solution of its density-score equations.
8. Stack the selected mediator and outcome scores with $\mathbf{X}_i - \bar{\mathbf{X}}$, form the empirical sandwich, and apply the analytic gradient in (S1).

The log-variance parameterization guarantees positive variance without post-solution truncation. Score evaluation additionally uses $\kappa_n = \min(10^{-10}, n^{-2})$ for the density floor and positive- Q_2 check stated in the main article. At every sample size in the reported analyses, $(a_n, b_n, d_n, \kappa_n) = (10^{-9}, 10^{-6}, 10^{-4}, 10^{-10})$, so the asymptotic sequences reproduce the retained numerical results exactly. Root acceptance requires score convergence, Jacobian invertibility and a defined covariance, with no empirically chosen coefficient or variance bounds. These checks establish numerical validity of the returned solution; Proposition S1 provides the corresponding first-order statistical link.

S5 Error distributions and complete simulation results

S5.1 Design and comparator specifications

All figures in the article and this supplement were generated with base R graphics.

The original simulation random-number seeds are set in `R/run_simulations.R`. The main experiment uses seed 20260827, the comparator experiment 20260828, the size-power experiment 20260829, and the algorithm-sensitivity experiment 20260830. Each task receives a stored task-specific seed before parallel execution, so the results do not depend on the number of forked workers. The independently calibrated power experiment has separate seeds and retained replication-level output, as detailed in Section S6.

The comparator experiment uses $n = 300$, 1,000 replications per distribution, 1,000 quasi-Bayesian draws and 499 bootstrap resamples, as specified in the main article. The file `comparator_seeds.csv` records the data-set seeds; `R/run_comparator.R` supports independent execution and resumable checkpoints.

For clarity about the comparator specification, the linked linear model implies

$$\mathbb{E}\{Y(t, M(t')) \mid \mathbf{X} = \mathbf{x}\} = \alpha_3 + \beta_3 t + (\gamma + \eta t)(\alpha_2 + \beta_2 t' + \boldsymbol{\xi}_2^\top \mathbf{x}) + \boldsymbol{\xi}_3^\top \mathbf{x}.$$

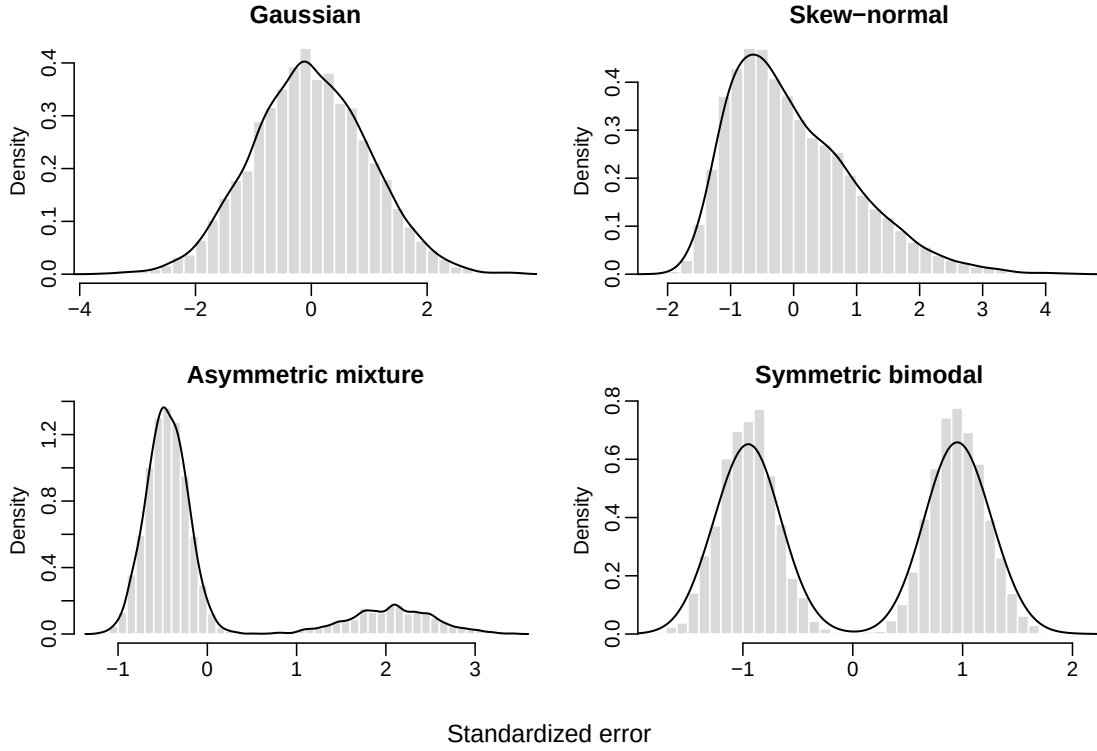


Figure S1: Monte Carlo views of the four standardized error distributions. Each panel uses 5,000 draws; the histogram is normalized to integrate to one and the line is a kernel density estimate

Thus the compatible natural effect model includes $t\mathbf{x}$ as well as tt' . In both the comparator simulation and the application, `medflex` uses outcome imputation with $T*M$ and every TX_j term, and a natural effect model with T_0*T_1 and every T_0X_j interaction. The extra TX_j terms ensure that the imputation specification passes the package's working-model compatibility check; their true coefficients are zero in the simulation. This gives the comparator a slightly more flexible outcome working model while preserving the same causal target. Continuous and binary baseline covariates are centered at their empirical means. The package's decomposition is evaluated at zero on that centered scale, which gives the plug-in population average under this linear structure. Its native standard errors condition on the covariate reference, whereas the stacked estimator adds the baseline-mean influence term.

S5.2 Precision and interval calibration

Tables S1 and S2 extend the main article's representative results to all effects and distributions, including bias, coverage, interval length and numerical success. Section S5.3 gives the common-data comparison.

Table S1: Complete main Monte Carlo results; Reps. is the number of numerically successful fits used for performance summaries, whereas Success uses all attempts; errors: G, Gaussian; SN, skew-normal; AM, asymmetric mixture; BM, symmetric bimodal

n	Error	Method	Effect	Reps.	Bias	RMSE	Cov.	MCSE	Length	Success
200	G	OLS	PNIE	1000	0.003	0.121	0.947	0.007	0.467	1.000
200	G	OLS	TNIE	1000	0.001	0.053	0.925	0.008	0.197	1.000
200	G	OLS	PNDE	1000	-0.002	0.178	0.950	0.007	0.698	1.000
200	G	OLS	TNDE	1000	-0.003	0.176	0.946	0.007	0.699	1.000
200	G	OLS	TE	1000	-0.001	0.162	0.949	0.007	0.643	1.000
200	G	Proposed	PNIE	1000	0.004	0.130	0.932	0.008	0.495	1.000
200	G	Proposed	TNIE	1000	0.001	0.056	0.912	0.009	0.210	1.000
200	G	Proposed	PNDE	1000	-0.002	0.191	0.934	0.008	0.742	1.000
200	G	Proposed	TNDE	1000	-0.005	0.189	0.941	0.007	0.740	1.000
200	G	Proposed	TE	1000	-0.001	0.175	0.938	0.008	0.682	1.000
200	SN	OLS	PNIE	1000	0.005	0.122	0.945	0.007	0.469	1.000
200	SN	OLS	TNIE	1000	0.001	0.051	0.911	0.009	0.194	1.000
200	SN	OLS	PNDE	1000	-0.001	0.185	0.936	0.008	0.701	1.000
200	SN	OLS	TNDE	1000	-0.005	0.181	0.951	0.007	0.701	1.000
200	SN	OLS	TE	1000	0.000	0.171	0.937	0.008	0.645	1.000
200	SN	Proposed	PNIE	1000	0.005	0.100	0.951	0.007	0.388	1.000
200	SN	Proposed	TNIE	1000	0.002	0.043	0.918	0.009	0.163	1.000
200	SN	Proposed	PNDE	1000	-0.001	0.155	0.941	0.007	0.594	1.000
200	SN	Proposed	TNDE	1000	-0.004	0.152	0.949	0.007	0.597	1.000
200	SN	Proposed	TE	1000	0.000	0.144	0.936	0.008	0.553	1.000
200	AM	OLS	PNIE	1000	0.001	0.126	0.942	0.007	0.475	1.000
200	AM	OLS	TNIE	1000	0.000	0.050	0.895	0.010	0.190	1.000
200	AM	OLS	PNDE	1000	0.009	0.183	0.948	0.007	0.702	1.000
200	AM	OLS	TNDE	1000	0.008	0.186	0.933	0.008	0.699	1.000
200	AM	OLS	TE	1000	0.008	0.168	0.946	0.007	0.646	1.000
200	AM	Proposed	PNIE	887	-0.001	0.055	0.940	0.008	0.120	0.887
200	AM	Proposed	TNIE	887	0.000	0.021	0.939	0.008	0.050	0.887
200	AM	Proposed	PNDE	887	0.002	0.093	0.935	0.008	0.332	0.887
200	AM	Proposed	TNDE	887	0.004	0.093	0.933	0.008	0.333	0.887
200	AM	Proposed	TE	887	0.002	0.088	0.937	0.008	0.327	0.887
200	BM	OLS	PNIE	1000	0.004	0.119	0.940	0.008	0.470	1.000
200	BM	OLS	TNIE	1000	-0.002	0.050	0.902	0.009	0.193	1.000
200	BM	OLS	PNDE	1000	0.005	0.169	0.953	0.007	0.700	1.000
200	BM	OLS	TNDE	1000	-0.002	0.177	0.944	0.007	0.698	1.000
200	BM	OLS	TE	1000	0.002	0.157	0.957	0.006	0.646	1.000
200	BM	Proposed	PNIE	921	0.000	0.034	0.929	0.008	0.121	0.921
200	BM	Proposed	TNIE	921	0.000	0.031	0.928	0.008	0.050	0.921
200	BM	Proposed	PNDE	921	0.002	0.120	0.948	0.007	0.335	0.921
200	BM	Proposed	TNDE	921	0.002	0.114	0.934	0.008	0.334	0.921
200	BM	Proposed	TE	921	0.002	0.110	0.950	0.007	0.327	0.921

n	Error	Method	Effect	Reps.	Bias	RMSE	Cov.	MCSE	Length	Success
500	G	OLS	PNIE	1000	0.001	0.074	0.956	0.006	0.297	1.000
500	G	OLS	TNIE	1000	0.000	0.031	0.941	0.007	0.122	1.000
500	G	OLS	PNDE	1000	-0.006	0.114	0.958	0.006	0.444	1.000
500	G	OLS	TNDE	1000	-0.007	0.117	0.931	0.008	0.443	1.000
500	G	OLS	TE	1000	-0.006	0.107	0.946	0.007	0.409	1.000
500	G	Proposed	PNIE	1000	0.001	0.078	0.958	0.006	0.311	1.000
500	G	Proposed	TNIE	1000	0.001	0.033	0.930	0.008	0.127	1.000
500	G	Proposed	PNDE	1000	-0.006	0.119	0.953	0.007	0.462	1.000
500	G	Proposed	TNDE	1000	-0.006	0.122	0.934	0.008	0.461	1.000
500	G	Proposed	TE	1000	-0.005	0.111	0.950	0.007	0.424	1.000
500	SN	OLS	PNIE	1000	0.000	0.077	0.955	0.007	0.298	1.000
500	SN	OLS	TNIE	1000	0.001	0.031	0.930	0.008	0.122	1.000
500	SN	OLS	PNDE	1000	-0.003	0.116	0.941	0.007	0.444	1.000
500	SN	OLS	TNDE	1000	-0.002	0.118	0.942	0.007	0.444	1.000
500	SN	OLS	TE	1000	-0.001	0.109	0.948	0.007	0.409	1.000
500	SN	Proposed	PNIE	1000	0.004	0.059	0.942	0.007	0.233	1.000
500	SN	Proposed	TNIE	1000	0.000	0.024	0.938	0.008	0.095	1.000
500	SN	Proposed	PNDE	1000	-0.001	0.093	0.943	0.007	0.361	1.000
500	SN	Proposed	TNDE	1000	-0.005	0.094	0.944	0.007	0.361	1.000
500	SN	Proposed	TE	1000	-0.001	0.087	0.944	0.007	0.337	1.000
500	AM	OLS	PNIE	1000	-0.004	0.075	0.946	0.007	0.299	1.000
500	AM	OLS	TNIE	1000	0.000	0.031	0.928	0.008	0.120	1.000
500	AM	OLS	PNDE	1000	0.007	0.116	0.940	0.008	0.444	1.000
500	AM	OLS	TNDE	1000	0.012	0.111	0.956	0.006	0.443	1.000
500	AM	OLS	TE	1000	0.007	0.106	0.947	0.007	0.409	1.000
500	AM	Proposed	PNIE	999	0.000	0.019	0.949	0.007	0.074	0.999
500	AM	Proposed	TNIE	999	0.000	0.008	0.943	0.007	0.030	0.999
500	AM	Proposed	PNDE	999	0.002	0.055	0.948	0.007	0.209	0.999
500	AM	Proposed	TNDE	999	0.003	0.055	0.948	0.007	0.209	0.999
500	AM	Proposed	TE	999	0.002	0.054	0.948	0.007	0.205	0.999
500	BM	OLS	PNIE	1000	-0.003	0.076	0.945	0.007	0.298	1.000
500	BM	OLS	TNIE	1000	0.000	0.032	0.934	0.008	0.122	1.000
500	BM	OLS	PNDE	1000	-0.003	0.110	0.953	0.007	0.443	1.000
500	BM	OLS	TNDE	1000	0.000	0.109	0.963	0.006	0.444	1.000
500	BM	OLS	TE	1000	-0.003	0.101	0.955	0.007	0.409	1.000
500	BM	Proposed	PNIE	1000	0.000	0.019	0.959	0.006	0.076	1.000
500	BM	Proposed	TNIE	1000	0.000	0.008	0.939	0.008	0.031	1.000
500	BM	Proposed	PNDE	1000	-0.001	0.054	0.937	0.008	0.210	1.000
500	BM	Proposed	TNDE	1000	-0.001	0.053	0.955	0.007	0.211	1.000
500	BM	Proposed	TE	1000	-0.001	0.053	0.941	0.007	0.206	1.000

Table S2: Complete comparator Monte Carlo results based on 1,000 attempts per cell; Reps. is the number of numerically successful fits used for performance summaries, whereas Success uses all attempts; QB denotes quasi-Bayes and Boot denotes bootstrap; other error abbreviations are defined in Table S1

n	Error	Method	Effect	Reps.	Bias	RMSE	Cov.	MCSE	Length	Success
300	G	OLS	PNIE	1000	0.001	0.100	0.952	0.007	0.383	1.000
300	G	OLS	TNIE	1000	-0.001	0.040	0.927	0.008	0.158	1.000
300	G	OLS	PNDE	1000	-0.001	0.146	0.956	0.006	0.571	1.000
300	G	OLS	TNDE	1000	-0.003	0.145	0.944	0.007	0.573	1.000
300	G	OLS	TE	1000	-0.002	0.135	0.955	0.007	0.527	1.000
300	G	Imai-QB	PNIE	1000	0.001	0.100	0.950	0.007	0.387	1.000
300	G	Imai-QB	TNIE	1000	-0.001	0.040	0.944	0.007	0.162	1.000
300	G	Imai-QB	PNDE	1000	-0.001	0.146	0.964	0.006	0.613	1.000
300	G	Imai-QB	TNDE	1000	-0.003	0.145	0.958	0.006	0.615	1.000
300	G	Imai-QB	TE	1000	-0.002	0.135	0.962	0.006	0.573	1.000
300	G	Imai-Boot	PNIE	1000	0.001	0.100	0.946	0.007	0.384	1.000
300	G	Imai-Boot	TNIE	1000	-0.001	0.040	0.940	0.008	0.161	1.000
300	G	Imai-Boot	PNDE	1000	-0.003	0.157	0.963	0.006	0.608	1.000
300	G	Imai-Boot	TNDE	1000	-0.005	0.158	0.958	0.006	0.609	1.000
300	G	Imai-Boot	TE	1000	-0.004	0.147	0.966	0.006	0.565	1.000
300	G	NEM	PNIE	1000	0.001	0.100	0.950	0.007	0.385	1.000
300	G	NEM	TNIE	1000	-0.001	0.040	0.922	0.008	0.160	1.000
300	G	NEM	PNDE	1000	-0.001	0.146	0.951	0.007	0.567	1.000
300	G	NEM	TNDE	1000	-0.003	0.146	0.941	0.007	0.569	1.000
300	G	NEM	TE	1000	-0.002	0.135	0.954	0.007	0.522	1.000
300	G	Proposed	PNIE	1000	0.000	0.107	0.942	0.007	0.406	1.000
300	G	Proposed	TNIE	1000	-0.001	0.042	0.922	0.008	0.166	1.000
300	G	Proposed	PNDE	1000	-0.001	0.151	0.945	0.007	0.599	1.000
300	G	Proposed	TNDE	1000	-0.002	0.154	0.942	0.007	0.602	1.000
300	G	Proposed	TE	1000	-0.002	0.139	0.948	0.007	0.552	1.000
300	SN	OLS	PNIE	1000	0.004	0.099	0.950	0.007	0.385	1.000
300	SN	OLS	TNIE	1000	-0.001	0.042	0.904	0.009	0.156	1.000
300	SN	OLS	PNDE	1000	-0.007	0.143	0.954	0.007	0.572	1.000
300	SN	OLS	TNDE	1000	-0.012	0.141	0.951	0.007	0.572	1.000
300	SN	OLS	TE	1000	-0.008	0.133	0.952	0.007	0.527	1.000
300	SN	Imai-QB	PNIE	1000	0.004	0.100	0.949	0.007	0.388	1.000
300	SN	Imai-QB	TNIE	1000	-0.001	0.042	0.933	0.008	0.162	1.000
300	SN	Imai-QB	PNDE	1000	-0.007	0.143	0.966	0.006	0.613	1.000
300	SN	Imai-QB	TNDE	1000	-0.012	0.142	0.971	0.005	0.614	1.000
300	SN	Imai-QB	TE	1000	-0.008	0.133	0.970	0.005	0.572	1.000
300	SN	Imai-Boot	PNIE	1000	0.004	0.099	0.939	0.008	0.386	1.000
300	SN	Imai-Boot	TNIE	1000	-0.001	0.042	0.922	0.008	0.159	1.000
300	SN	Imai-Boot	PNDE	1000	-0.006	0.156	0.968	0.006	0.609	1.000
300	SN	Imai-Boot	TNDE	1000	-0.011	0.153	0.965	0.006	0.609	1.000
300	SN	Imai-Boot	TE	1000	-0.007	0.146	0.968	0.006	0.566	1.000
300	SN	NEM	PNIE	1000	0.004	0.099	0.950	0.007	0.386	1.000
300	SN	NEM	TNIE	1000	-0.001	0.043	0.901	0.009	0.158	1.000
300	SN	NEM	PNDE	1000	-0.007	0.143	0.948	0.007	0.568	1.000
300	SN	NEM	TNDE	1000	-0.012	0.141	0.954	0.007	0.568	1.000
300	SN	NEM	TE	1000	-0.008	0.133	0.950	0.007	0.522	1.000
300	SN	Proposed	PNIE	1000	0.002	0.079	0.955	0.007	0.310	1.000
300	SN	Proposed	TNIE	1000	0.001	0.033	0.924	0.008	0.127	1.000
300	SN	Proposed	PNDE	1000	-0.006	0.121	0.949	0.007	0.476	1.000
300	SN	Proposed	TNDE	1000	-0.007	0.117	0.959	0.006	0.477	1.000
300	SN	Proposed	TE	1000	-0.006	0.112	0.947	0.007	0.443	1.000

n	Error	Method	Effect	Reps.	Bias	RMSE	Cov.	MCSE	Length	Success
300	AM	OLS	PNIE	1000	0.000	0.101	0.944	0.007	0.387	1.000
300	AM	OLS	TNIE	1000	0.000	0.041	0.886	0.010	0.153	1.000
300	AM	OLS	PNDE	1000	-0.001	0.152	0.939	0.008	0.573	1.000
300	AM	OLS	TNDE	1000	-0.001	0.148	0.945	0.007	0.572	1.000
300	AM	OLS	TE	1000	-0.001	0.141	0.935	0.008	0.527	1.000
300	AM	Imai-QB	PNIE	1000	0.000	0.102	0.944	0.007	0.389	1.000
300	AM	Imai-QB	TNIE	1000	0.000	0.041	0.947	0.007	0.162	1.000
300	AM	Imai-QB	PNDE	1000	-0.001	0.152	0.958	0.006	0.615	1.000
300	AM	Imai-QB	TNDE	1000	-0.002	0.148	0.960	0.006	0.616	1.000
300	AM	Imai-QB	TE	1000	-0.001	0.141	0.954	0.007	0.573	1.000
300	AM	Imai-Boot	PNIE	1000	0.000	0.101	0.945	0.007	0.390	1.000
300	AM	Imai-Boot	TNIE	1000	0.000	0.041	0.920	0.009	0.154	1.000
300	AM	Imai-Boot	PNDE	1000	0.000	0.164	0.951	0.007	0.610	1.000
300	AM	Imai-Boot	TNDE	1000	0.000	0.159	0.964	0.006	0.611	1.000
300	AM	Imai-Boot	TE	1000	0.000	0.153	0.953	0.007	0.567	1.000
300	AM	NEM	PNIE	1000	0.001	0.102	0.948	0.007	0.388	1.000
300	AM	NEM	TNIE	1000	0.000	0.042	0.890	0.010	0.155	1.000
300	AM	NEM	PNDE	1000	-0.001	0.152	0.937	0.008	0.568	1.000
300	AM	NEM	TNDE	1000	-0.002	0.149	0.941	0.007	0.567	1.000
300	AM	NEM	TE	1000	-0.001	0.140	0.935	0.008	0.522	1.000
300	AM	Proposed	PNIE	962	0.001	0.026	0.938	0.008	0.096	0.962
300	AM	Proposed	TNIE	962	0.000	0.012	0.949	0.007	0.039	0.962
300	AM	Proposed	PNDE	962	0.000	0.070	0.945	0.007	0.270	0.962
300	AM	Proposed	TNDE	962	-0.001	0.071	0.948	0.007	0.270	0.962
300	AM	Proposed	TE	962	0.000	0.069	0.944	0.007	0.264	0.962
300	BM	OLS	PNIE	1000	0.002	0.096	0.956	0.006	0.385	1.000
300	BM	OLS	TNIE	1000	-0.002	0.038	0.928	0.008	0.157	1.000
300	BM	OLS	PNDE	1000	0.004	0.144	0.948	0.007	0.571	1.000
300	BM	OLS	TNDE	1000	0.001	0.138	0.952	0.007	0.572	1.000
300	BM	OLS	TE	1000	0.002	0.133	0.951	0.007	0.527	1.000
300	BM	Imai-QB	PNIE	1000	0.002	0.096	0.956	0.006	0.388	1.000
300	BM	Imai-QB	TNIE	1000	-0.002	0.038	0.950	0.007	0.161	1.000
300	BM	Imai-QB	PNDE	1000	0.004	0.144	0.958	0.006	0.615	1.000
300	BM	Imai-QB	TNDE	1000	0.000	0.138	0.965	0.006	0.616	1.000
300	BM	Imai-QB	TE	1000	0.002	0.134	0.960	0.006	0.574	1.000
300	BM	Imai-Boot	PNIE	1000	0.002	0.096	0.957	0.006	0.386	1.000
300	BM	Imai-Boot	TNIE	1000	-0.002	0.038	0.949	0.007	0.160	1.000
300	BM	Imai-Boot	PNDE	1000	0.003	0.158	0.961	0.006	0.608	1.000
300	BM	Imai-Boot	TNDE	1000	-0.001	0.153	0.967	0.006	0.608	1.000
300	BM	Imai-Boot	TE	1000	0.001	0.148	0.961	0.006	0.566	1.000
300	BM	NEM	PNIE	1000	0.002	0.096	0.955	0.007	0.386	1.000
300	BM	NEM	TNIE	1000	-0.002	0.038	0.929	0.008	0.159	1.000
300	BM	NEM	PNDE	1000	0.004	0.144	0.948	0.007	0.567	1.000
300	BM	NEM	TNDE	1000	0.000	0.138	0.951	0.007	0.568	1.000
300	BM	NEM	TE	1000	0.002	0.134	0.948	0.007	0.522	1.000
300	BM	Proposed	PNIE	978	-0.001	0.029	0.949	0.007	0.098	0.978
300	BM	Proposed	TNIE	978	0.000	0.020	0.943	0.007	0.040	0.978
300	BM	Proposed	PNDE	978	-0.001	0.080	0.943	0.007	0.272	0.978
300	BM	Proposed	TNDE	978	0.001	0.080	0.943	0.007	0.272	0.978
300	BM	Proposed	TE	978	-0.001	0.079	0.941	0.008	0.266	0.978

S5.3 Paired-fit and reporting reliability

The script `R/paired_fit_audit.R` joins OLS and proposed records by study, sample size, distribution, replication and effect, then recomputes RMSE and interval-length ratios on common successful fits. All OLS fits succeeded, so the paired count equals the proposed-method successful-fit count.

For each method we calculate conditional coverage among the paired fits. For the proposal we additionally report

$$\frac{\#\{\text{attempts producing a valid interval covering the truth}\}}{\#\{\text{all attempted data sets}\}}.$$

This report-and-cover fraction equals success rate times conditional coverage. It measures the reliability of the complete numerical procedure and complements nominal coverage conditional on a returned interval.

The full CSV output contains all 60 study-by-design-by-effect rows. For $R = \sqrt{\bar{u}/\bar{v}}$, where u_i and v_i are the paired squared estimation errors of the proposed and OLS methods, the reported approximate Monte Carlo standard error is

$$\widehat{\text{MCSE}}(R) = \frac{R}{\sqrt{N}} \text{sd} \left\{ \frac{1}{2} \left(\frac{u_i}{\bar{u}} - \frac{v_i}{\bar{v}} \right) \right\}.$$

This paired delta-method MCSE is conditional on retained fits; all-attempted summaries separately represent fit availability.

Table S3: Paired-fit audit using identical data sets for OLS and the proposal. Main and comparator designs each have 1,000 attempts per cell. N is the number of paired valid fits. Ratios use proposed over OLS; C_O and C_P are conditional coverages. $R\&C$ is the fraction of all attempts yielding a proposed interval that covers the truth, not conventional conditional coverage. G, SN, AM and BM denote the four error distributions. All five effects and ratio Monte Carlo standard errors are in the accompanying CSV file.

Study	n	Error	Effect	N	RMSE ratio	Length ratio	C_O	C_P	$R\&C$
Main	200	G	PNIE	1000	1.069	1.060	0.947	0.932	0.932
Main	200	G	TE	1000	1.078	1.061	0.949	0.938	0.938
Main	200	SN	PNIE	1000	0.820	0.828	0.945	0.951	0.951
Main	200	SN	TE	1000	0.840	0.857	0.937	0.936	0.936
Main	200	AM	PNIE	887	0.444	0.251	0.949	0.940	0.834
Main	200	AM	TE	887	0.550	0.504	0.966	0.937	0.831
Main	200	BM	PNIE	921	0.289	0.258	0.942	0.929	0.856
Main	200	BM	TE	921	0.731	0.506	0.966	0.950	0.875
Main	500	G	PNIE	1000	1.044	1.048	0.956	0.958	0.958
Main	500	G	TE	1000	1.035	1.037	0.946	0.950	0.950
Main	500	SN	PNIE	1000	0.761	0.783	0.955	0.942	0.942
Main	500	SN	TE	1000	0.802	0.822	0.948	0.944	0.944
Main	500	AM	PNIE	999	0.255	0.247	0.947	0.949	0.948
Main	500	AM	TE	999	0.507	0.500	0.947	0.948	0.947
Main	500	BM	PNIE	1000	0.247	0.255	0.945	0.959	0.959
Main	500	BM	TE	1000	0.519	0.503	0.955	0.941	0.941
Comparator	300	G	PNIE	1000	1.071	1.058	0.952	0.942	0.942
Comparator	300	G	TE	1000	1.032	1.048	0.955	0.948	0.948
Comparator	300	SN	PNIE	1000	0.797	0.807	0.950	0.955	0.955
Comparator	300	SN	TE	1000	0.846	0.840	0.952	0.947	0.947
Comparator	300	AM	PNIE	962	0.261	0.249	0.948	0.938	0.902
Comparator	300	AM	TE	962	0.506	0.501	0.943	0.944	0.908
Comparator	300	BM	PNIE	978	0.303	0.256	0.955	0.949	0.928
Comparator	300	BM	TE	978	0.598	0.505	0.954	0.941	0.920

S6 Null size, power and numerical sensitivity

S6.1 Original nominal comparison

Table S4 gives the original nominal comparison at $n = 300$ with 1,000 attempts per effect value. The independently calibrated comparison below separates detection gains from differences in attained null size.

Table S4: Empirical size and power for PNIE under the standardized asymmetric-mixture error, $n = 300$, based on 1,000 attempted replications. Rejection and MCSE are conditional on numerically successful fits; Success uses all attempts

β_2	True PNIE	Method	Reps.	Success	Reject	MCSE
0.00	0.000	OLS delta	1000	1.000	0.016	0.004
0.00	0.000	Proposed	1000	0.964	0.056	0.007
0.10	-0.026	OLS delta	1000	1.000	0.052	0.007
0.10	-0.026	Proposed	1000	0.968	0.946	0.007
0.20	-0.052	OLS delta	1000	1.000	0.195	0.013
0.20	-0.052	Proposed	1000	0.962	1.000	0.000
0.30	-0.078	OLS delta	1000	1.000	0.431	0.016
0.30	-0.078	Proposed	1000	0.958	1.000	0.000
0.40	-0.104	OLS delta	1000	1.000	0.653	0.015
0.40	-0.104	Proposed	1000	0.960	1.000	0.000

S6.2 Independent null calibration and Huber comparison

The asymmetric-mixture design uses $n = 300$, $\gamma = -0.26$, $\eta = 0.8$ and $\beta_2 \in \{0, 0.1, 0.2, 0.3, 0.4\}$, giving $\text{PNIE} = -0.26\beta_2$. All other parameters and the OLS and proposed estimators are unchanged. The design was recorded in `POWER_CALIBRATION_PLAN_20260923.md`; all four methods analyze common data within each replication.

Huber target and implementation

The comparators are motivated by Wang et al. (2025), whose data-driven Huber tuning uses a preliminary least-absolute-deviation (LAD) regression. We fit the same mediator and outcome formulas as the main analysis, $M \sim T + X$ and $Y \sim T * M + X$, using `quantreg::rq` with quantile 0.5 and method `br` for the pilot. For each equation, let r_i^{LAD} be the pilot residual and $s = \text{median}(|r_i^{\text{LAD}}|)/0.6745$. Huber-FIX sets the raw residual threshold to $k = 1.345s$. Huber-SEL minimizes

$$\hat{\tau}(k) = \frac{n^{-1} \sum_i \min\{(r_i^{\text{LAD}})^2, k^2\}}{\{n^{-1} \sum_i \mathbf{1}(|r_i^{\text{LAD}}| \leq k)\}^2}$$

over $k = 0.20, 0.21, \dots, \text{round}(3s, 2)$, following the grid in the authors' public implementation. A zero denominator is assigned infinite objective; an empty grid is a recorded failure. For the chosen threshold, iteratively reweighted least squares minimizes the convex Huber loss, starting from the LAD fit. It stops when the largest coefficient change is below $10^{-10}(1 + \|\hat{b}\|_\infty)$, with a limit of 500 iterations and an additional absolute mean-score tolerance of 10^{-7} .

Both comparators target the same PNIE despite the asymmetric errors. To see this, write either regression as $R = \alpha + Z^\top b + \epsilon$, where Z contains all nonconstant regressors, and put $\psi_k(u) = \max\{-k, \min(k, u)\}$. Independence of ϵ and Z implies that the population Huber equations are solved by slope b and intercept $\alpha + d_k$, where $\mathbb{E}\{\psi_k(\epsilon - d_k)\} = 0$. Thus the intercept absorbs a constant location shift and the slopes retain their structural meanings. Consequently, the product of the mediator's treatment slope and the outcome's mediator slope targets $\beta_2\gamma$. The experiment uses this slope-only target, rather than interpreting Huber intercepts as conditional-mean intercepts for direct or total effects.

For the covariance calculation define $C = \text{Var}(Z)$ and $B_k = \mathbb{E}\{\mathbf{1}(|\epsilon - d_k| < k)\}$. Under the common independent-error law and usual nonsingularity and smoothness conditions, the slope influence function is

$$\text{IF}_b = C^{-1}(Z - \mathbb{E}Z) \frac{\psi_k(\epsilon - d_k)}{B_k}.$$

The centered-design slope equations have zero population derivative with respect to the intercept and the threshold: the relevant derivatives factor into $\mathbb{E}(Z - \mathbb{E}Z) = 0$ and an error-only expectation. This orthogonality explains why estimating a convergent pilot-based threshold does not contribute a first-order slope term under this model. We evaluate the empirical influence function at the *final Huber residuals*, using the centered empirical design covariance and the fitted inlier fraction. After centering the observation-level influences, PNIE has estimated influence

$$\widehat{\text{IF}}_{\text{PNIE},i} = \widehat{\gamma}_H \widehat{\text{IF}}_{\beta_2,i} + \widehat{\beta}_{2,H} \widehat{\text{IF}}_{\gamma,i}, \quad \widehat{\text{SE}}_H = \frac{\{\sum_i \widehat{\text{IF}}_{\text{PNIE},i}^2\}^{1/2}}{n}.$$

This includes the empirical cross-regression covariance. It is an interaction-adapted comparison, rather than a verbatim reproduction of the simple-mediation Sobel implementation in Wang et al. (2025). Evaluating the score variance at the fitted Huber location also distinguishes this calculation from a pilot-residual variance approximation when errors are asymmetric.

Independent calibration and evaluation

For method j , let $Z_{jr} = |\widehat{\text{PNIE}}_{jr} / \widehat{\text{SE}}_{jr}|$. We first generate 5,000 independent null data sets at $\beta_2 = 0$. Among the m_j numerically valid null fits, order the statistics and select

$$c_j = Z_{j,(\lceil 0.95(m_j+1) \rceil)}.$$

We then freeze these four critical values and evaluate $\mathbf{1}(Z_{jr} > c_j)$ on 1,000 *new* data sets at each of the five effect values. The held-out null is disjoint from calibration and provides the reported type I error check. The nominal comparison on these same evaluation data uses $z_{0.975}$ for all methods. Master seed 20260923 generates 10,000 distinct task seeds, retained with every fit status and numerical result. The null-calibration target is 5%; its held-out estimate need not be exactly 0.05 because both stages have Monte Carlo variation. These DGP-specific critical values diagnose whether the original power difference can be attributed to unequal null size. They are not proposed as data-analysis critical values for an unknown error law or as uniform calibration over the composite mediation null. In particular, γ is fixed away from zero here, so the joint-zero nonregular case is outside this experiment.

Conditional rejection uses each method's valid-fit denominator. Report-and-reject divides the number of usable rejections by all 1,000 attempts. We additionally compare methods on the common subset where all four fits succeed. These matched comparisons retain the method-specific critical values obtained from calibration. For simulation uncertainty, 1,000 bootstrap replications (seed 20260924) independently resample calibration tasks and evaluation tasks, retaining method pairing within each resample, and recompute the thresholds and rejection rates. The same resampled calibration thresholds are used across the five effect settings within a bootstrap replication. The reported two-stage MCSE is the standard deviation across these replications, and intervals are pointwise percentile intervals for Monte Carlo uncertainty. Binomial MCSEs conditional on the realized thresholds are also retained in the project repository. At cells with no observed failures to reject or no observed paired disagreement, empirical MCSEs and bootstrap intervals can be zero-width; these describe the retained simulation outcomes rather than asserting exact unit power or identical population rejection probabilities.

Tables S5–S7 give the calibration cutoffs, held-out rejection rates and common-data contrasts, respectively. They retain both conditional power and all-attempt detection for the weak- and larger-effect comparisons interpreted in the main article.

Table S5: Independent null-calibration results from 5,000 attempted data sets per method. The interval for the critical value reflects 1,000 resamples of the calibration tasks.

Method	Valid	Critical value	Monte Carlo 95% interval
OLS	5000	1.669	[1.636, 1.712]
Huber-FIX	5000	1.938	[1.885, 1.982]
Huber-SEL	5000	1.988	[1.929, 2.045]
Proposed	4841	1.967	[1.926, 2.026]

Table S6: Held-out PNIE rejection probabilities. Each effect uses 1,000 new data sets. Nominal and calibrated rates condition on valid fits; Report/reject uses all attempts. MCSE includes both calibration and evaluation uncertainty. At $\beta_2 = 0$ the rates estimate type I error, and elsewhere they estimate power.

β_2	Method	Valid	Nominal	Calibrated	MCSE	Report/reject
0.0	OLS	1000	0.018	0.054	0.008	0.054
0.0	Huber-FIX	1000	0.048	0.057	0.010	0.057
0.0	Huber-SEL	1000	0.055	0.052	0.008	0.052
0.0	Proposed	965	0.055	0.052	0.008	0.050
0.1	OLS	1000	0.043	0.112	0.012	0.112
0.1	Huber-FIX	1000	0.644	0.652	0.018	0.652
0.1	Huber-SEL	1000	0.730	0.721	0.018	0.721
0.1	Proposed	962	0.928	0.927	0.009	0.892
0.2	OLS	1000	0.174	0.323	0.018	0.323
0.2	Huber-FIX	1000	0.990	0.992	0.003	0.992
0.2	Huber-SEL	1000	0.999	0.999	0.001	0.999
0.2	Proposed	960	1.000	1.000	0.000	0.960
0.3	OLS	1000	0.473	0.632	0.018	0.632
0.3	Huber-FIX	1000	1.000	1.000	0.000	1.000
0.3	Huber-SEL	1000	1.000	1.000	0.000	1.000
0.3	Proposed	967	1.000	1.000	0.000	0.967
0.4	OLS	1000	0.631	0.789	0.015	0.789
0.4	Huber-FIX	1000	1.000	1.000	0.000	1.000
0.4	Huber-SEL	1000	1.000	1.000	0.000	1.000
0.4	Proposed	967	1.000	1.000	0.000	0.967

Table S7: Calibrated rejection contrasts on data sets where all four methods succeed. Difference is proposed minus comparator. The Monte Carlo interval uses 1,000 paired, two-stage bootstrap resamples and describes simulation uncertainty, not uncertainty in an individual mediation analysis.

β_2	Comparator	Common valid	Difference	MCSE	95% interval
0.0	OLS	965	0.000	0.012	[-0.019, 0.026]
0.0	Huber-FIX	965	-0.003	0.011	[-0.021, 0.024]
0.0	Huber-SEL	965	0.000	0.010	[-0.018, 0.020]
0.1	OLS	962	0.814	0.015	[0.782, 0.842]
0.1	Huber-FIX	962	0.269	0.018	[0.236, 0.301]
0.1	Huber-SEL	962	0.204	0.017	[0.172, 0.238]
0.2	OLS	960	0.677	0.019	[0.642, 0.714]
0.2	Huber-FIX	960	0.006	0.003	[0.002, 0.013]
0.2	Huber-SEL	960	0.001	0.001	[0.000, 0.003]
0.3	OLS	967	0.364	0.018	[0.327, 0.401]
0.3	Huber-FIX	967	0.000	0.000	[0.000, 0.000]
0.3	Huber-SEL	967	0.000	0.000	[0.000, 0.000]
0.4	OLS	967	0.206	0.015	[0.177, 0.239]
0.4	Huber-FIX	967	0.000	0.000	[0.000, 0.000]
0.4	Huber-SEL	967	0.000	0.000	[0.000, 0.000]

The complete records, selected Huber thresholds, warning and failure messages, calibration cutoffs, paired summaries and two-stage bootstrap output are retained in the project repository. Independent checks reconstruct data without the study’s generator wrapper, reproduce all four methods on ten prespecified audit tasks, and verify all reported counts and summary denominators. Estimator-level tests compare Huber IRLS with an independent optimizer, check response-shift invariance, and verify that a sufficiently large threshold reproduces OLS coefficients and slope covariance.

S6.3 Starting-value sensitivity

Table S8 varies the coefficient perturbation scale using 300 common data sets per mixture at $n = 200$. It records valid-fit estimates and success rates separately; unsuccessful density-score fits remain failures.

Table S8: Sensitivity to the coefficient perturbation scale at $n = 200$, based on 300 common data sets per error distribution. Success uses all attempts; other summaries use numerically successful fits

Error	Scale	Reps.	Success	Mean PNIE	SD	Roots M	Roots Y
Asymmetric mixture	0.05	300	0.957	-0.320	0.037	1.01	1.00
Asymmetric mixture	0.10	300	0.900	-0.319	0.037	1.00	1.01
Asymmetric mixture	0.20	300	0.860	-0.321	0.029	1.00	1.01
Symmetric bimodal	0.05	300	0.950	-0.319	0.029	1.00	1.01
Symmetric bimodal	0.10	300	0.940	-0.319	0.029	1.00	1.04
Symmetric bimodal	0.20	300	0.910	-0.318	0.075	1.00	1.09

S7 Variance-misspecification sensitivity

This supplementary experiment retains the main-study conditional means,

$$M = 0.2 + 0.4T + 0.3X + s(T, X)U_M, \quad Y = 0.5T - 0.8M + TM + 0.4X + s(T, X)U_Y.$$

Here $X \sim N(0, 1)$, $T \sim \text{Bernoulli}(0.5)$, and the standardized innovations U_M, U_Y are independent of one another and of (T, X) . We use the Gaussian and asymmetric-mixture densities in Section S5, with no change to their component parameters. Each error has conditional mean zero. For covariate-dependent variance,

$$s_X^2(X) = \exp(\kappa X - \kappa^2/2),$$

and for treatment-dependent variance,

$$s_T^2(T) = \exp\{\kappa(2T - 1)\} / \cosh(\kappa).$$

Both have population expectation one. Normalization uses the known population law, not the realized sample. For $\kappa = 0.15, 0.30, 0.60$, the covariate mechanism's variance ratio between the 75th and 25th percentiles of X is approximately 1.22, 1.50, 2.25; the treatment-to-control variance ratio is 1.35, 1.82, 3.32. These ratios describe different contrasts and should not be treated as identical strengths of heterogeneity.

To see why causal identification is preserved in this design, define $M(t)$ by replacing T with t in the mediator equation and define $Y(t, m)$ by replacing (T, M) with (t, m) in the outcome equation. Conditional on X , the two potential variables depend on independent innovations; randomized T is independent of both. This structural construction supplies sequential ignorability, which is stronger than conditional mean-zero restrictions alone. For positive κ , the observed errors can depend on the regressors through their scales. The population effects nevertheless remain $\text{PNIE} = -0.32$, $\text{TNIE} = 0.08$, $\text{PNDE} = 0.70$, $\text{TNDE} = 1.10$ and $\text{TE} = 0.78$, because the structural conditional means are unchanged.

We use $n = 300$ and 500 attempts per innovation-by-variance configuration. Sharing the constant-variance reference gives seven configurations per density and 7,000 data sets. Seed 20260917 generates the stored task seeds. Configurations within a replication share X, T, U_M, U_Y ; replications are independent. We compare OLS with its stacked empirical sandwich against the unchanged proposed estimator, retaining every fit status, including failures.

Table S9 reports PNIE and TE; the project repository retains all five effects. Estimation and interval metrics use successful fits. Bias MCSE is the empirical standard deviation divided by the square root of the successful-fit count; coverage MCSE is the binomial standard error with that denominator. Coverage 0.95 with 500 successful fits has MCSE 0.0097. Numerical success and report-and-cover use all 500 attempts; paired summaries use common successful replications.

Figure S2 displays the mechanism-specific coverage patterns discussed in the main article. Table S9 adds bias, RMSE, interval length and success rates, allowing coverage changes to be read with estimation error and fit availability.

Table S9: Variance-misspecification sensitivity at $n = 300$, with 500 attempted replications per configuration. Coverage and precision are conditional on numerical success. Bias and coverage MCSEs are in parentheses; Success uses all attempts.

Error	Scale	κ	Effect	Method	Bias (MCSE)	RMSE	Coverage (MCSE)	Length	Success	R+C
AM	None	0.00	PNIE	OLS	-0.004 (0.004)	0.097	0.954 (0.009)	0.388	1.000	0.954
AM	None	0.00	PNIE	Semi	-0.002 (0.001)	0.028	0.952 (0.010)	0.097	0.968	0.922
AM	None	0.00	TE	OLS	0.003 (0.006)	0.129	0.950 (0.010)	0.527	1.000	0.950
AM	None	0.00	TE	Semi	0.001 (0.003)	0.070	0.950 (0.010)	0.266	0.968	0.920
AM	X	0.15	PNIE	OLS	-0.004 (0.004)	0.097	0.954 (0.009)	0.388	1.000	0.954
AM	X	0.15	PNIE	Semi	-0.002 (0.001)	0.026	0.959 (0.009)	0.098	0.976	0.936
AM	X	0.15	TE	OLS	0.003 (0.006)	0.129	0.956 (0.009)	0.527	1.000	0.956
AM	X	0.15	TE	Semi	-0.000 (0.003)	0.066	0.955 (0.009)	0.266	0.976	0.932
AM	X	0.30	PNIE	OLS	-0.004 (0.004)	0.097	0.954 (0.009)	0.389	1.000	0.954
AM	X	0.30	PNIE	Semi	-0.002 (0.001)	0.026	0.963 (0.009)	0.100	0.978	0.942
AM	X	0.30	TE	OLS	0.002 (0.006)	0.129	0.958 (0.009)	0.527	1.000	0.958
AM	X	0.30	TE	Semi	0.000 (0.003)	0.065	0.959 (0.009)	0.266	0.978	0.938
AM	X	0.60	PNIE	OLS	-0.004 (0.004)	0.100	0.950 (0.010)	0.394	1.000	0.950
AM	X	0.60	PNIE	Semi	-0.002 (0.001)	0.027	0.948 (0.010)	0.106	0.920	0.872
AM	X	0.60	TE	OLS	0.002 (0.006)	0.129	0.960 (0.009)	0.525	1.000	0.960
AM	X	0.60	TE	Semi	0.000 (0.003)	0.067	0.957 (0.010)	0.269	0.920	0.880
AM	T	0.15	PNIE	OLS	-0.004 (0.004)	0.097	0.952 (0.010)	0.387	1.000	0.952
AM	T	0.15	PNIE	Semi	0.026 (0.001)	0.036	0.802 (0.018)	0.099	0.972	0.780
AM	T	0.15	TE	OLS	0.002 (0.006)	0.127	0.956 (0.009)	0.518	1.000	0.956
AM	T	0.15	TE	Semi	-0.025 (0.003)	0.069	0.922 (0.012)	0.261	0.972	0.896
AM	T	0.30	PNIE	OLS	-0.004 (0.004)	0.098	0.954 (0.009)	0.387	1.000	0.954
AM	T	0.30	PNIE	Semi	0.065 (0.001)	0.070	0.327 (0.021)	0.103	0.972	0.318
AM	T	0.30	TE	OLS	0.002 (0.006)	0.124	0.958 (0.009)	0.510	1.000	0.958
AM	T	0.30	TE	Semi	-0.054 (0.004)	0.112	0.844 (0.016)	0.262	0.972	0.820
AM	T	0.60	PNIE	OLS	-0.005 (0.004)	0.099	0.952 (0.010)	0.387	1.000	0.952
AM	T	0.60	PNIE	Semi	0.180 (0.001)	0.182	0.000 (0.000)	0.109	0.798	0.000
AM	T	0.60	TE	OLS	0.001 (0.005)	0.120	0.960 (0.009)	0.494	1.000	0.960
AM	T	0.60	TE	Semi	-0.157 (0.013)	0.306	0.393 (0.024)	0.277	0.798	0.314
G	None	0.00	PNIE	OLS	-0.001 (0.005)	0.102	0.948 (0.010)	0.385	1.000	0.948
G	None	0.00	PNIE	Semi	0.002 (0.005)	0.105	0.932 (0.011)	0.408	1.000	0.932
G	None	0.00	TE	OLS	0.003 (0.006)	0.135	0.952 (0.010)	0.527	1.000	0.952
G	None	0.00	TE	Semi	0.003 (0.006)	0.141	0.934 (0.011)	0.549	1.000	0.934
G	X	0.15	PNIE	OLS	-0.001 (0.005)	0.102	0.950 (0.010)	0.385	1.000	0.950
G	X	0.15	PNIE	Semi	0.001 (0.005)	0.105	0.944 (0.010)	0.406	1.000	0.944
G	X	0.15	TE	OLS	0.003 (0.006)	0.135	0.950 (0.010)	0.526	1.000	0.950
G	X	0.15	TE	Semi	0.003 (0.006)	0.143	0.932 (0.011)	0.548	1.000	0.932
G	X	0.30	PNIE	OLS	-0.001 (0.005)	0.102	0.946 (0.010)	0.386	1.000	0.946
G	X	0.30	PNIE	Semi	0.000 (0.005)	0.106	0.934 (0.011)	0.411	1.000	0.934
G	X	0.30	TE	OLS	0.003 (0.006)	0.136	0.942 (0.010)	0.526	1.000	0.942
G	X	0.30	TE	Semi	0.004 (0.006)	0.145	0.930 (0.011)	0.555	1.000	0.930
G	X	0.60	PNIE	OLS	-0.001 (0.005)	0.104	0.938 (0.011)	0.390	1.000	0.938
G	X	0.60	PNIE	Semi	-0.000 (0.005)	0.109	0.938 (0.011)	0.416	1.000	0.938
G	X	0.60	TE	OLS	0.003 (0.006)	0.137	0.932 (0.011)	0.525	1.000	0.932
G	X	0.60	TE	Semi	0.006 (0.006)	0.145	0.946 (0.010)	0.558	1.000	0.946
G	T	0.15	PNIE	OLS	-0.000 (0.005)	0.102	0.946 (0.010)	0.385	1.000	0.946
G	T	0.15	PNIE	Semi	0.001 (0.005)	0.107	0.944 (0.010)	0.420	1.000	0.944
G	T	0.15	TE	OLS	0.003 (0.006)	0.132	0.952 (0.010)	0.518	1.000	0.952
G	T	0.15	TE	Semi	0.003 (0.006)	0.139	0.952 (0.010)	0.558	1.000	0.952
G	T	0.30	PNIE	OLS	0.000 (0.005)	0.102	0.942 (0.010)	0.385	1.000	0.942
G	T	0.30	PNIE	Semi	0.001 (0.005)	0.111	0.950 (0.010)	0.455	1.000	0.950
G	T	0.30	TE	OLS	0.002 (0.006)	0.130	0.952 (0.010)	0.509	1.000	0.952
G	T	0.30	TE	Semi	0.001 (0.006)	0.142	0.952 (0.010)	0.599	1.000	0.952
G	T	0.60	PNIE	OLS	0.001 (0.005)	0.102	0.940 (0.011)	0.386	1.000	0.940
G	T	0.60	PNIE	Semi	0.003 (0.006)	0.132	0.952 (0.010)	0.553	1.000	0.952
G	T	0.60	TE	OLS	0.001 (0.006)	0.125	0.960 (0.009)	0.494	1.000	0.960
G	T	0.60	TE	Semi	-0.000 (0.007)	0.165	0.954 (0.009)	0.716	1.000	0.954

G: Gaussian; AM: asymmetric mixture; Semi: the unchanged proposed estimator. R+C is the fraction of all attempted data sets yielding an interval covering the truth, not conventional coverage. Scale X denotes covariate-dependent variance and scale T treatment-dependent variance; None is their common constant-variance reference. All five effects and paired-fit summaries are retained in the project repository.

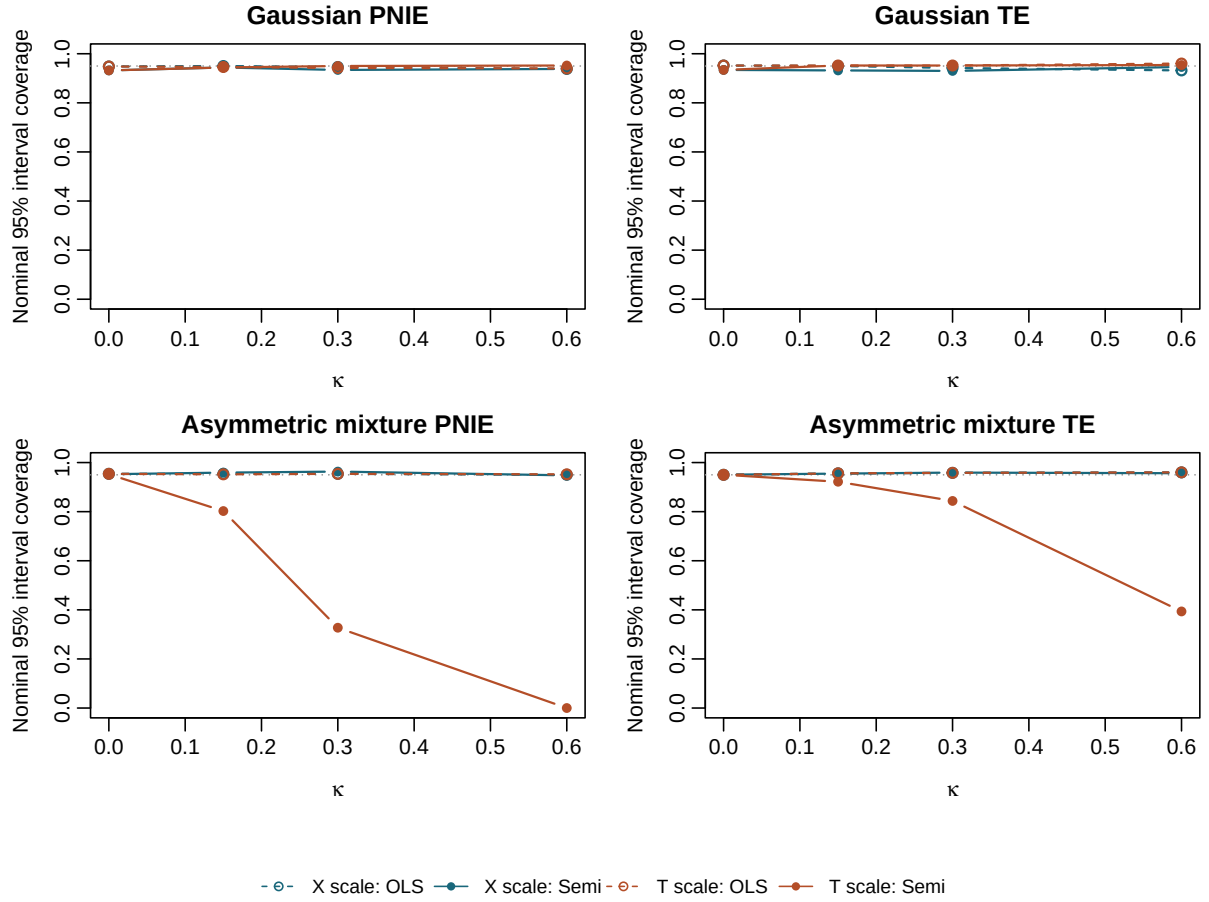


Figure S2: Coverage conditional on numerical success in the variance-misspecification experiment. Each point summarizes 500 attempted replications; the shared constant-variance reference is at $\kappa = 0$. Colors distinguish covariate- and treatment-dependent scales, and line types distinguish OLS from the original proposed estimator. The dotted horizontal line marks the nominal 0.95 reference; Table S9 reports coverage MCSEs and numerical success.

S8 Reduced-form confounding-sensitivity extension

S8.1 Identification and attribution

This section develops the density-score estimation and covariance propagation for the fixed-error-correlation sensitivity identity of Imai, Keele and Yamamoto (2010), as cited in the main article. Let $m_t(\mathbf{X}) = \alpha_2 + \beta_2 t + \boldsymbol{\xi}_2^\top \mathbf{X}$ and $q_t(\mathbf{X}) = a_t + \mathbf{d}_t^\top \mathbf{X}$. A sufficient structural formulation is

$$M(t) = m_t(\mathbf{X}) + U_t, \quad Y(t, m) = q_t(\mathbf{X}) + b_t\{m - m_t(\mathbf{X})\} + E_t.$$

Assume treatment exchangeability conditional on baseline $\mathbf{X}$, consistency and positivity; U_t, E_t have conditional mean zero; and E_t is invariant to the intervention value m . The sensitivity model allows mediator–outcome dependence through the joint law of U_t and E_t . The observed reduced-form error is $W_t = b_t U_t + E_t$. Its conditional variance and covariance with U_t obey

$$\omega_t = b_t v_M + \rho_t \sqrt{v_M \text{Var}(E_t)}, \quad v_t = b_t^2 v_M + \text{Var}(E_t) + 2b_t \rho_t \sqrt{v_M \text{Var}(E_t)}.$$

Solving with positive $\text{Var}(E_t)$ yields the main article’s sensitivity slope. The natural indirect effect is $b_t \mathbb{E}\{M(1) - M(0)\} = b_t \beta_2$, since the additive E_t cancels in the difference of nested potential outcomes. The total effect is $\mathbb{E}(q_1 - q_0)$, giving the direct effects by the two decomposition identities. Thus the linear structural form supplies the causal interpretation and defines the sensitivity class: additive mediator–outcome confounding with an intervention-invariant structural error and a stratum-specific constant mediator slope.

For estimation, $U = M - m_T(\mathbf{X})$ has a common law independent of $(T, \mathbf{X})$, and the joint law of (U, W_t) is stable across $\mathbf{X}$ within each stratum. These assumptions support pooled-mediator and stratum-specific reduced-outcome fits with unspecified marginal densities. The figures follow the common- ρ path through (ρ_0, ρ_1) .

S8.2 Estimating equations and covariance

Let S_M be the marginal regression score for M on $(1, T, \mathbf{X}^\top)$ and S_{W_t} the score for Y on $(1, \mathbf{X}^\top)$ within $T = t$. The stacked population function is

$$\Psi_i(\theta) = \begin{pmatrix} S_{M,i} \\ 1(T_i = 0)S_{W_0,i} \\ 1(T_i = 1)S_{W_1,i} \\ 1(T_i = 0)(U_i W_{0i} - \omega_0) \\ 1(T_i = 1)(U_i W_{1i} - \omega_1) \\ \mathbf{X}_i - \boldsymbol{\mu}_X \end{pmatrix}.$$

The first three blocks use the density-score equations and numerical settings in the main article. The OLS benchmark replaces each regression block by $(De, e^2/v - 1)$, with parameters $(\beta, \log v)$, so both methods estimate the same coefficient and variance nuisance parameters. Cross-covariances are the sample averages of fitted residual products within treatment groups. The parameter domain requires $v_M v_t - \omega_t^2 > 0$; fitted values outside this domain are recorded transparently as numerical failures rather than altered by truncation.

Let $\hat{A} = n^{-1} \sum_i \partial \hat{\Psi}_i / \partial \theta^\top$ and $\hat{B} = n^{-1} \sum_i \hat{\Psi}_i \hat{\Psi}_i^\top$. Within-stratum regression bread blocks are multiplied by n_t/n . The cross-product row for stratum t has derivatives

$$-n^{-1} \sum_{i:T_i=t} D_{Mi} \hat{W}_{ti}, \quad -n^{-1} \sum_{i:T_i=t} D_{Wi} \hat{U}_i,$$

with respect to the mediator and corresponding reduced-outcome coefficients, respectively, and derivative $-n_t/n$ with respect to ω_t . It has zero direct derivative with respect to log variances. The baseline-mean block has bread $-I$. Empirical off-diagonal bread entries are retained even where their population expectations vanish. The full meat includes cross-covariance between the marginal regression scores and residual-product moments. The implemented covariance is $\hat{A}^{-1} \hat{B} \hat{A}^{-\top} / n$.

Conditional asymptotic justification. Assume positive treatment probabilities and nonsingular bread. Also require finite score second moments and fourth error moments, and $v_M v_t > \omega_t^2$. Each fitted marginal regression is assumed to satisfy the marginal analogues of Conditions (R1)–(R4) in Section S3. Require covariance consistency as in Condition (R5) for the full augmented stack, retaining its cross-block dependence rather than imposing the independent-error factorization used for the original mediation model. On any fixed finite collection of sensitivity values strictly inside $(-1, 1)$, the marginal expansions hold jointly. Taylor expansion of the appended residual-product equations and empirical baseline mean then gives the stacked expansion $\varphi_\theta = -A^{-1}\Psi$. Applying the chain rule gives $\varphi_{g,\rho} = G_\rho \varphi_\theta$ and the displayed sandwich covariance. Positive effect variance is required for a scalar Wald interval. Thus standard estimating-equation and delta-method theory supplies pointwise asymptotic inference under the stated marginal-estimator regularity. The efficiency interpretation is attached to the marginal score blocks; the displayed joint representation characterizes the assembled estimator in the reduced-form working model.

S8.3 Analytic gradients and zero boundaries

Put $\ell_t = \omega_t/v_M$, $A_t = v_t/v_M$, $d_t = \sqrt{A_t - \ell_t^2}$ and $h_t = \rho_t/\sqrt{1 - \rho_t^2}$. Then $b_t = \ell_t - h_t d_t$ and

$$\begin{aligned}\frac{\partial b_t}{\partial \omega_t} &= \frac{1 + h_t \ell_t / d_t}{v_M}, \\ \frac{\partial b_t}{\partial \log v_M} &= -\ell_t + \frac{h_t (A_t - 2\ell_t^2)}{2d_t}, \\ \frac{\partial b_t}{\partial \log v_t} &= -\frac{h_t A_t}{2d_t}.\end{aligned}$$

Thus $d\delta_t = b_t d\beta_2 + \beta_2 db_t$. For the total effect,

$$d\tau = d(a_1 - a_0) + \boldsymbol{\mu}_X^\top d(\mathbf{d}_1 - \mathbf{d}_0) + (\mathbf{d}_1 - \mathbf{d}_0)^\top d\boldsymbol{\mu}_X.$$

Direct-effect gradients are obtained by subtraction. The code supports separate (ρ_0, ρ_1) even though the figures use their common-value path.

For $\beta_2 \neq 0$, $\delta_t(\rho_t) = 0$ exactly at $r_t = \omega_t/\sqrt{v_M v_t}$. Its derivatives are $\partial r_t/\partial \omega_t = 1/\sqrt{v_M v_t}$ and $\partial r_t/\partial \log v_M = \partial r_t/\partial \log v_t = -r_t/2$. We form a delta interval for $\text{atanh}(r_t)$ and transform its endpoints back with $\tanh$. At $\beta_2 = 0$, the complete indirect-effect curve equals zero, producing a set-valued rather than unique effect-zero threshold. The confidence interval for a unique zero boundary addresses uncertainty in the point-estimate threshold, while the statistical-significance boundary along the curve addresses pointwise interval exclusion of zero.

Under the working error independence from regressors, the first-order derivatives of residual second moments with respect to the mean coefficients vanish. Their influence terms reduce to $U^2 - v_M$, $1(T = t)(W_t^2 - v_t)/p_t$ and $1(T = t)(UW_t - \omega_t)/p_t$. Consequently the OLS-RF and DS-RF zero-boundary estimators share the same first-order residual-moment representation under this model, explaining their close finite-sample boundaries. Density-score gains for the indirect-effect curve arise instead through the pooled mediator treatment coefficient and its joint covariance with the remaining nuisance estimates.

For the contour calibration, Imai’s additional shared-confounder variance decomposition expresses U and E_t as linear contributions of a hypothetical pretreatment confounder plus uncorrelated remainder errors, with the corresponding orthogonality to the confounder. It gives $\rho_t^2 = R_M^{2*} R_{Y,t}^{2*}$, with the sign specified separately. The outcome fraction refers to the *structural outcome error*, which is distinct from the observed reduced-form outcome residual. The figure uses this moment calibration to map hypothetical latent variance fractions into sensitivity values without requiring a complete latent distribution. Accordingly, its axes are labeled as residual-variance fractions rather than total-variance tilde quantities.

S8.4 Simulation design and verification

The main article’s confounding experiment uses seed 20260918, $n = 300$ and 500 replications per innovation-by-correlation cell. For independent verification, the stratum- t covariance is $-0.8 + t + \rho_{\text{true}}$ and the reduced-outcome variance is $(-0.8 + t + \rho_{\text{true}})^2 + 1 - \rho_{\text{true}}^2$; these determine the population zero boundaries. The original natural effects are retained. Evaluation at the generating correlation and at zero includes valid-fit, paired-fit and all-attempted summaries, with every fit status recorded.

Table S10 gives performance at the generating correlation; the full-curve and zero-boundary assessments follow below. JOBS II curves are grouped with the application in Section S9.

Table S10: Reduced-form sensitivity estimation at the true structural correlation: $n = 300, 500$ attempts per cell. Coverage and precision use valid fits; Success uses all attempts.

Error	ρ	Effect	Method	Bias	RMSE	Coverage (MCSE)	Length	Success
AM	-0.3	PNIE	OLS-RF	0.017	0.067	0.966 (0.008)	0.257	1.000
AM	-0.3	PNIE	DS-RF	0.000	0.066	0.958 (0.009)	0.260	0.996
AM	-0.3	TE	OLS-RF	-0.011	0.145	0.950 (0.010)	0.560	1.000
AM	-0.3	TE	DS-RF	-0.011	0.145	0.946 (0.010)	0.560	0.996
AM	0.0	PNIE	OLS-RF	0.003	0.070	0.946 (0.010)	0.283	1.000
AM	0.0	PNIE	DS-RF	0.002	0.049	0.952 (0.010)	0.198	1.000
AM	0.0	TE	OLS-RF	-0.004	0.128	0.964 (0.008)	0.527	1.000
AM	0.0	TE	DS-RF	-0.004	0.127	0.968 (0.008)	0.527	1.000
AM	0.3	PNIE	OLS-RF	0.011	0.079	0.974 (0.007)	0.324	1.000
AM	0.3	PNIE	DS-RF	0.001	0.044	0.946 (0.010)	0.170	0.998
AM	0.3	TE	OLS-RF	-0.002	0.124	0.960 (0.009)	0.490	1.000
AM	0.3	TE	DS-RF	-0.002	0.124	0.958 (0.009)	0.490	0.998
G	-0.3	PNIE	OLS-RF	0.004	0.108	0.926 (0.012)	0.418	1.000
G	-0.3	PNIE	DS-RF	0.005	0.110	0.924 (0.012)	0.434	1.000
G	-0.3	TE	OLS-RF	0.000	0.146	0.932 (0.011)	0.558	1.000
G	-0.3	TE	DS-RF	0.000	0.146	0.934 (0.011)	0.558	1.000
G	0.0	PNIE	OLS-RF	0.004	0.102	0.946 (0.010)	0.397	1.000
G	0.0	PNIE	DS-RF	0.003	0.105	0.946 (0.010)	0.413	1.000
G	0.0	TE	OLS-RF	0.001	0.140	0.940 (0.011)	0.529	1.000
G	0.0	TE	DS-RF	0.001	0.140	0.942 (0.010)	0.529	1.000
G	0.3	PNIE	OLS-RF	0.005	0.098	0.940 (0.011)	0.385	1.000
G	0.3	PNIE	DS-RF	0.005	0.102	0.938 (0.011)	0.402	1.000
G	0.3	TE	OLS-RF	-0.000	0.119	0.962 (0.009)	0.490	1.000
G	0.3	TE	DS-RF	-0.000	0.119	0.962 (0.009)	0.491	1.000

G: Gaussian; AM: asymmetric mixture. OLS-RF and DS-RF use the same reduced-form models with OLS and marginal density-score regression, respectively. They are not the conditional-outcome estimators in the original simulation. All five effects, zero-rho analyses, threshold inference and paired comparisons are retained in the project repository.

The independent tests compare analytic and numerical gradients for all five effects, verify both exact decompositions and covariance symmetry, reconstruct the Imai covariance identity at known population parameters, and check zero-boundary derivatives and separate-correlation inputs. Output checks independently recompute performance summaries, compare paired replication identifiers, check all attempted/valid denominators, and reconstruct retained OLS fits from the recorded seeds. The existing core estimator is unchanged.

S8.5 Full-curve Monte Carlo assessment

The full-curve evaluation reconstructs the same 3,000 data sets and estimator settings from the stored seeds, as recorded in `CURVE_SIMULATION_PLAN_20260916.md`. Each fit supplies five effects on $\{-0.8, -0.75, \dots, 0.8\}$. With generating correlation r , the moments $v_M = 1$, $\omega_t = -0.8 + t + r$ and $v_t = \omega_t^2 + 1 - r^2$ give the main article’s population curve. The total effect is 0.78 throughout; direct effects follow by subtraction.

For assumed ρ , bias, RMSE and coverage target the population sensitivity functional; **CausalBias** and **CausalInclusion** target the generating causal effect. Paired ratios use common valid replication identifiers. Monte Carlo calculations preserve within-curve dependence and use the valid-fit or all-attempt denominators defined in Section S5.3.

For a common assumed correlation, write $h = \rho/\sqrt{1 - \rho^2}$. Each fitted effect has the exact form

$$\hat{g}(\rho) = \hat{A} + h\hat{B}, \quad \widehat{\text{Var}}\{\hat{g}(\rho)\} = \hat{V}_A + 2h\hat{C}_{AB} + h^2\hat{V}_B.$$

Here $A = g(\theta; 0)$ and $B = g(\theta; 1/\sqrt{2}) - g(\theta; 0)$, with gradients obtained from the previously displayed effect map. The covariance terms follow jointly from the complete nuisance sandwich. The project repository stores these five coefficients for every effect and attempted fit, allowing all grid values and intervals to be reconstructed exactly for any plotting grid without refitting, smoothing or interpolating confidence limits. Unsuccessful fits retain missing coefficients and a diagnostic message.

The main figure’s central 90% empirical ranges describe the sampling distribution of the effect estimates across valid fits. Figure S3 separately evaluates each fit’s nominal 95% pointwise interval, and Figure S4 compares RMSE on paired valid data sets. Reading the RMSE ratios together with pointwise coverage is particularly informative where the estimated curve is near zero. Table S11 adds a distinct evaluation of the estimated point-effect zero boundary.

Table S11: Zero-boundary inference in the confounding simulation. The same retained data sets are used; these are not additional independent replications. Coverage and length refer to the Fisher-transform interval for $\rho_t^\dagger$.

Error	ρ_{true}	Effect	Method	Bias	RMSE	Coverage (MCSE)	Length
AM	-0.3	PNIE	OLS-RF	0.003	0.068	0.958 (0.009)	0.276
AM	-0.3	PNIE	DS-RF	0.004	0.068	0.960 (0.009)	0.275
AM	-0.3	TNIE	OLS-RF	0.001	0.078	0.940 (0.011)	0.311
AM	-0.3	TNIE	DS-RF	0.002	0.077	0.944 (0.010)	0.311
AM	0.0	PNIE	OLS-RF	0.006	0.072	0.958 (0.009)	0.290
AM	0.0	PNIE	DS-RF	0.006	0.072	0.960 (0.009)	0.290
AM	0.0	TNIE	OLS-RF	0.008	0.081	0.946 (0.010)	0.311
AM	0.0	TNIE	DS-RF	0.007	0.081	0.944 (0.010)	0.312
AM	0.3	PNIE	OLS-RF	-0.002	0.079	0.950 (0.010)	0.306
AM	0.3	PNIE	DS-RF	-0.002	0.079	0.950 (0.010)	0.305
AM	0.3	TNIE	OLS-RF	-0.003	0.080	0.936 (0.011)	0.301
AM	0.3	TNIE	DS-RF	-0.004	0.079	0.936 (0.011)	0.301
G	-0.3	PNIE	OLS-RF	0.002	0.058	0.950 (0.010)	0.215
G	-0.3	PNIE	DS-RF	0.002	0.058	0.948 (0.010)	0.215
G	-0.3	TNIE	OLS-RF	0.002	0.080	0.954 (0.009)	0.311
G	-0.3	TNIE	DS-RF	0.002	0.080	0.952 (0.010)	0.311
G	0.0	PNIE	OLS-RF	0.006	0.063	0.946 (0.010)	0.243
G	0.0	PNIE	DS-RF	0.006	0.063	0.946 (0.010)	0.243
G	0.0	TNIE	OLS-RF	0.005	0.081	0.944 (0.010)	0.307
G	0.0	TNIE	DS-RF	0.005	0.081	0.948 (0.010)	0.307
G	0.3	PNIE	OLS-RF	0.008	0.069	0.954 (0.009)	0.274
G	0.3	PNIE	DS-RF	0.008	0.069	0.954 (0.009)	0.274
G	0.3	TNIE	OLS-RF	-0.000	0.069	0.946 (0.010)	0.274
G	0.3	TNIE	DS-RF	-0.000	0.069	0.944 (0.010)	0.274

G: Gaussian; AM: asymmetric mixture. Each cell starts with 500 attempts. Success denominators and failure records are the same as for Table S10. A zero boundary of the point-estimate curve is not an interval-zero boundary.

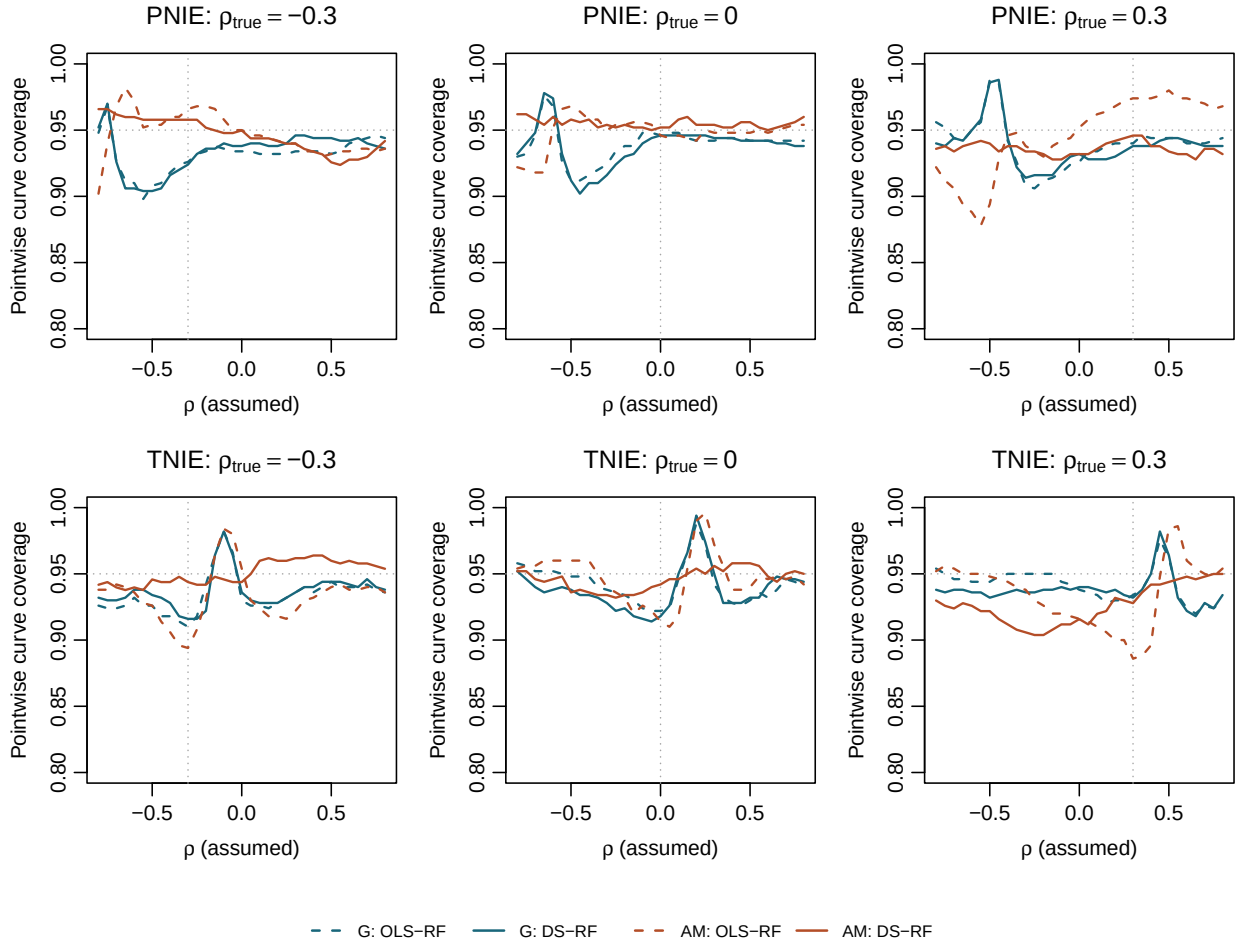


Figure S3: Pointwise coverage of the population sensitivity functional across the full assumed-correlation grid, conditional on numerical success. Each configuration starts with 500 attempts. G denotes Gaussian innovations and AM asymmetric-mixture innovations. The horizontal line is 0.95 and the vertical line is the generating correlation. At other correlations, coverage targets the corresponding sensitivity functional; at the generating correlation, that functional equals the generating causal effect. Complete Monte Carlo standard errors and report-and-cover fractions are retained in the project repository.

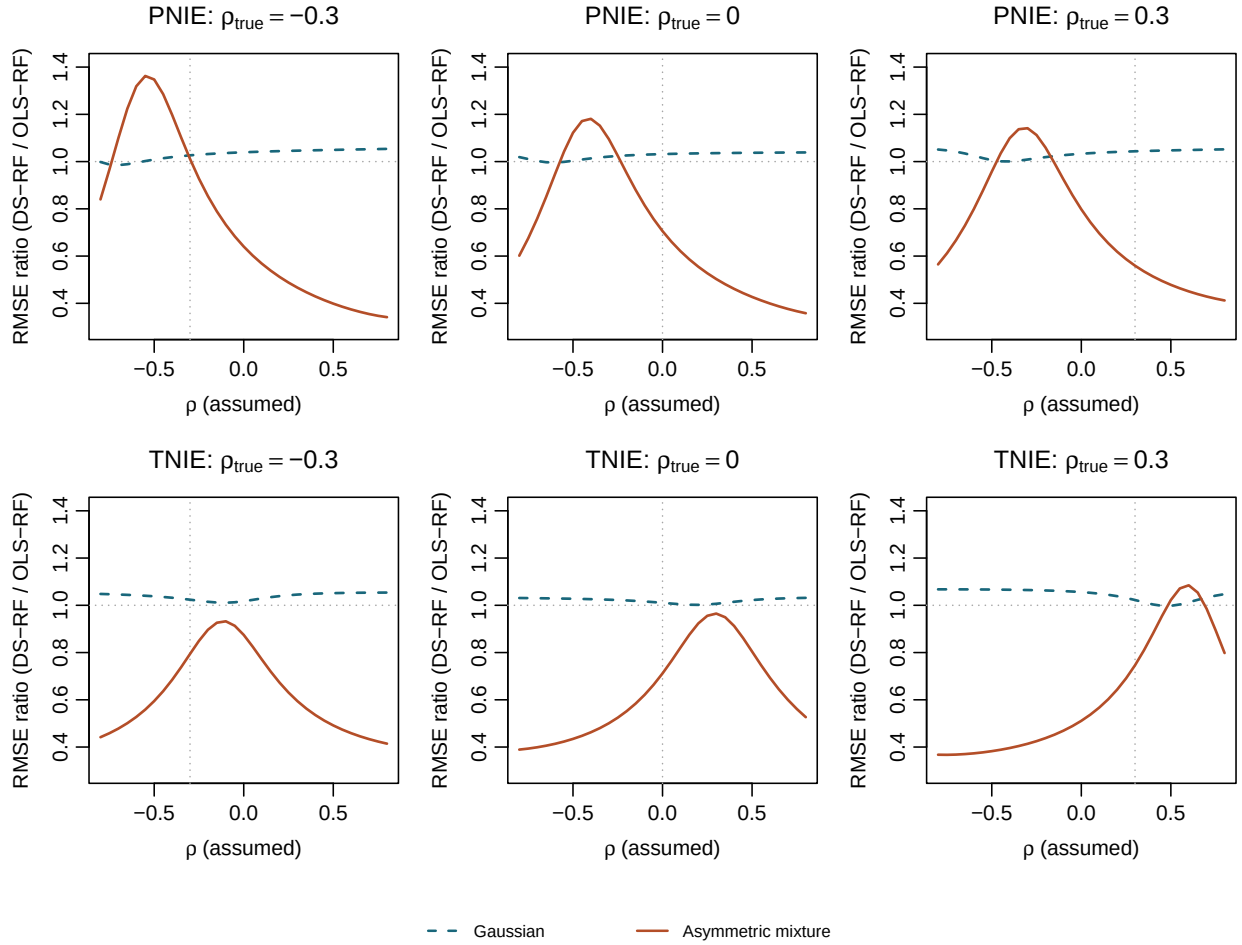


Figure S4: Paired DS-RF/OLS-RF RMSE ratios for the population PNIE/TNIE sensitivity curves. Ratios below one favor DS-RF in RMSE. The same successful data sets are used in the numerator and denominator, and all assumed correlations and generating configurations are retained. Pointwise coverage and numerical success provide the complementary calibration and reliability measures.

S9 Additional JOBS II results

The `jobs` object in R package `mediation` version 4.5.1 supplies all 899 complete observations used in the main article. The mediator `job_seek` has 25 distinct values on $[1, 5]$; the outcome `depress2` has 50 on $[1, 4.909]$. All analyses retain the main article’s treatment–mediator interaction and four-covariate adjustment unless stated otherwise.

S9.1 Natural effects and regression estimates

Table S12 gives the complete five-method comparison underlying the main article’s application figure; Table S13 records the linked-regression fits. The diagnostic checks below accompany these working-model estimates.

Table S12: Complete JOBS II natural-effect estimates and nominal 95% limits for five methods ($n = 899$). Negative values indicate lower post-intervention depressive symptoms under assignment to the intervention

Effect	Method	Estimate	Lower	Upper
PNIE	OLS delta	-0.0136	-0.0360	0.0087
PNIE	Imai quasi-Bayes	-0.0138	-0.0391	0.0080
PNIE	Imai bootstrap	-0.0136	-0.0373	0.0077
PNIE	<code>medflex</code> NEM	-0.0136	-0.0358	0.0087
PNIE	Proposed	-0.0109	-0.0205	-0.0013
TNIE	OLS delta	-0.0097	-0.0261	0.0067
TNIE	Imai quasi-Bayes	-0.0098	-0.0272	0.0060
TNIE	Imai bootstrap	-0.0097	-0.0289	0.0052
TNIE	<code>medflex</code> NEM	-0.0098	-0.0263	0.0068
TNIE	Proposed	-0.0072	-0.0144	-0.0001
PNDE	OLS delta	-0.0365	-0.1167	0.0437
PNDE	Imai quasi-Bayes	-0.0357	-0.1155	0.0451
PNDE	Imai bootstrap	-0.0367	-0.1150	0.0451
PNDE	<code>medflex</code> NEM	-0.0369	-0.1162	0.0425
PNDE	Proposed	-0.0272	-0.0892	0.0349
TNDE	OLS delta	-0.0326	-0.1116	0.0464
TNDE	Imai quasi-Bayes	-0.0317	-0.1128	0.0483
TNDE	Imai bootstrap	-0.0327	-0.1102	0.0473
TNDE	<code>medflex</code> NEM	-0.0330	-0.1113	0.0452
TNDE	Proposed	-0.0235	-0.0835	0.0364
TE	OLS delta	-0.0462	-0.1281	0.0357
TE	Imai quasi-Bayes	-0.0454	-0.1278	0.0370
TE	Imai bootstrap	-0.0464	-0.1278	0.0357
TE	<code>medflex</code> NEM	-0.0466	-0.1278	0.0346
TE	Proposed	-0.0344	-0.0962	0.0274

Table S13: Regression estimates for the JOBS II analysis. Variable names follow the reproducibility code

Equation	Method	Term	Estimate	SE
Mediator	OLS delta	(Intercept)	4.0172	0.1458
Mediator	OLS delta	T	0.0601	0.0490
Mediator	OLS delta	depress1	-0.2633	0.0462
Mediator	OLS delta	econ_hard	0.1026	0.0262
Mediator	OLS delta	sex	0.0241	0.0476
Mediator	OLS delta	age	0.0041	0.0024
Mediator	Proposed	(Intercept)	3.9798	0.0734
Mediator	Proposed	T	0.0725	0.0288
Mediator	Proposed	depress1	-0.1065	0.0274
Mediator	Proposed	econ_hard	0.0507	0.0146
Mediator	Proposed	sex	0.0248	0.0264
Mediator	Proposed	age	0.0013	0.0013
Outcome	OLS delta	(Intercept)	1.6210	0.2220
Outcome	OLS delta	T	-0.2998	0.2450
Outcome	OLS delta	M	-0.2269	0.0491
Outcome	OLS delta	depress1	0.4165	0.0363
Outcome	OLS delta	econ_hard	0.0684	0.0217
Outcome	OLS delta	sex	0.0566	0.0382
Outcome	OLS delta	age	0.0012	0.0017
Outcome	OLS delta	T:M	0.0658	0.0582
Outcome	Proposed	(Intercept)	1.7148	0.1553
Outcome	Proposed	T	-0.2286	0.1868
Outcome	Proposed	M	-0.1504	0.0331
Outcome	Proposed	depress1	0.2621	0.0334
Outcome	Proposed	econ_hard	0.0432	0.0152
Outcome	Proposed	sex	0.0369	0.0267
Outcome	Proposed	age	0.0003	0.0014
Outcome	Proposed	T:M	0.0504	0.0433

S9.2 Numerical and baseline-adjustment checks

Table S14 records estimates, starting-value choices, root counts and score norms across the three perturbation scales.

Table S14: JOBS II algorithm sensitivity across perturbation scales

Scale	PNIE	95% CI	Start M	Start Y	Roots M	Roots Y	Score M	Score Y
0.05	-0.011	[-0.020, -0.001]	1	1	1	1	2.05e-12	1.16e-12
0.10	-0.011	[-0.020, -0.001]	1	1	1	1	2.05e-12	1.16e-12
0.20	-0.011	[-0.020, -0.001]	1	1	1	1	2.05e-12	1.16e-12

Table S15 removes pretreatment depression from both regressions while retaining all other terms, observations and numerical settings. The companion file `application_covariate_sensitivity.csv` contains all five effects for both estimators. Baseline-depression adjustment remains primary because pretreatment symptoms are relevant to post-intervention depression and mediator–outcome confounding.

Table S15: JOBS II sensitivity to adjustment for pretreatment depressive symptoms. The sample, remaining covariates and treatment–mediator interaction are unchanged. The specification including baseline depression is primary; all five effects are retained in the accompanying CSV

Baseline depression	Method	Effect	Estimate	Lower	Upper
Included	OLS delta	PNIE	-0.0136	-0.0360	0.0087
Included	OLS delta	TE	-0.0462	-0.1281	0.0357
Included	Proposed	PNIE	-0.0109	-0.0205	-0.0013
Included	Proposed	TE	-0.0344	-0.0962	0.0274
Excluded	OLS delta	PNIE	-0.0198	-0.0495	0.0100
Excluded	OLS delta	TE	-0.0560	-0.1460	0.0340
Excluded	Proposed	PNIE	-0.0122	-0.0230	-0.0014
Excluded	Proposed	TE	-0.0328	-0.0964	0.0309

S9.3 Residual diagnostics and working-model interpretation

For the residual-spread diagnostic, squared OLS residuals are regressed on all non-intercept design columns. The statistic nR^2 uses a chi-squared reference with 5 and 7 degrees of freedom for the mediator and outcome models, respectively. Table S16 also reports skewness, excess kurtosis and one-dimensional spread tests using fitted values; Figure S5 visualizes density shape and spread.

Table S16: Baseline-adjusted OLS residual diagnostics. Normality p is from Shapiro–Wilk. Fitted p uses squared residuals regressed on fitted values; full LM is nR^2 using all original non-intercept design columns, with the displayed degrees of freedom (df). Tests are diagnostic checks, not tests of causal identification

Equation	Skew.	Ex. kurt.	Normality p	Fitted p	Full LM	df	Full p
Mediator	-0.781	0.856	1.81e-14	3.00e-02	13.06	5	2.28e-02
Outcome	1.083	1.875	1.04e-18	8.43e-08	30.50	7	7.67e-05

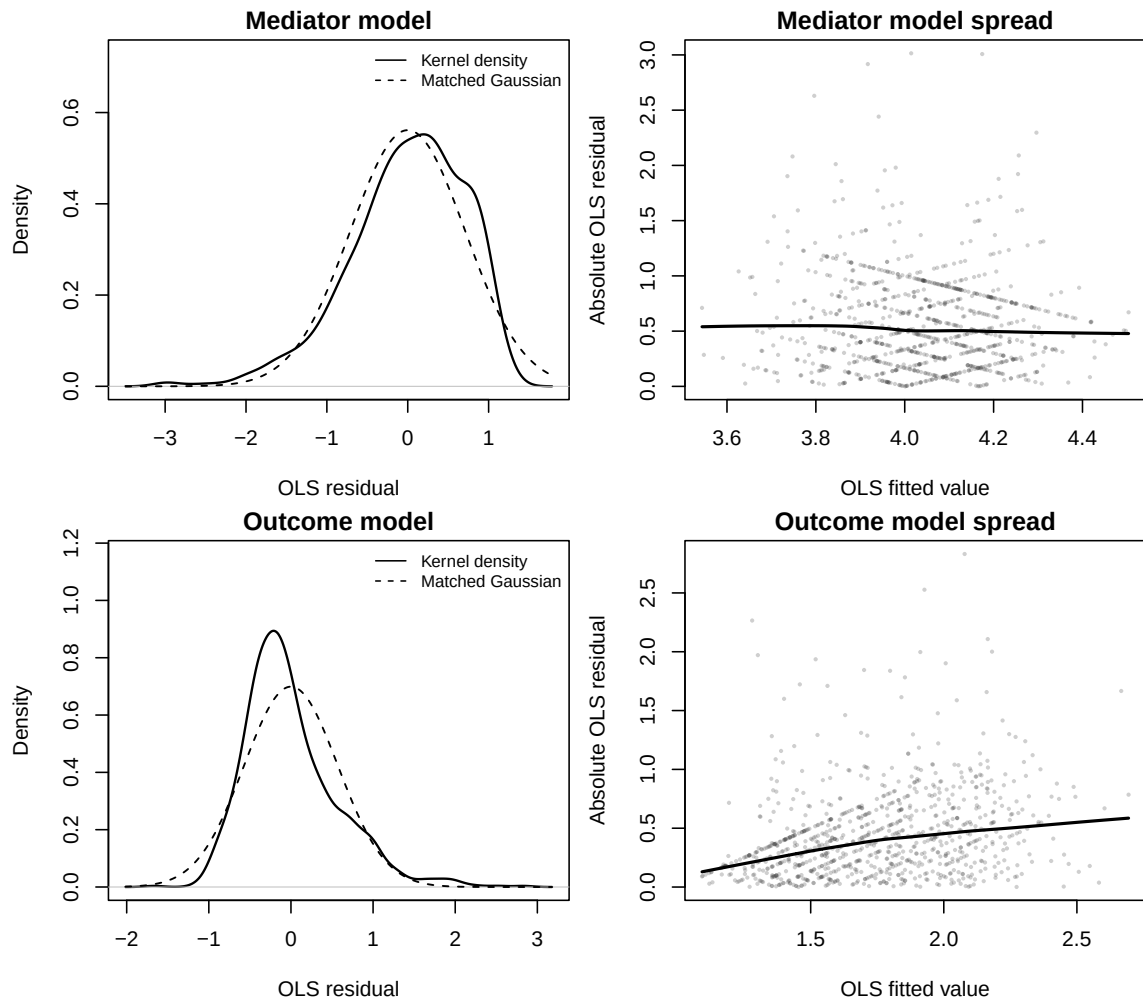


Figure S5: Baseline-adjusted OLS residual diagnostics for JOBS II. Left panels compare kernel residual densities with Gaussians having the same residual mean and standard deviation. Right panels plot absolute residuals against fitted values with lowess smooths; the upper and lower rows correspond to the mediator and outcome equations

S9.4 Confounding-sensitivity illustration

Figure S6 complements the main article’s contours with pointwise PNIE/TNIE curves on the same sample and adjustment set. Separate reduced-form diagnostics assess the marginal-error restrictions in Section S8. The project repository retains both signs of the residual- R^2 grid, zero-boundary intervals and regression diagnostics.

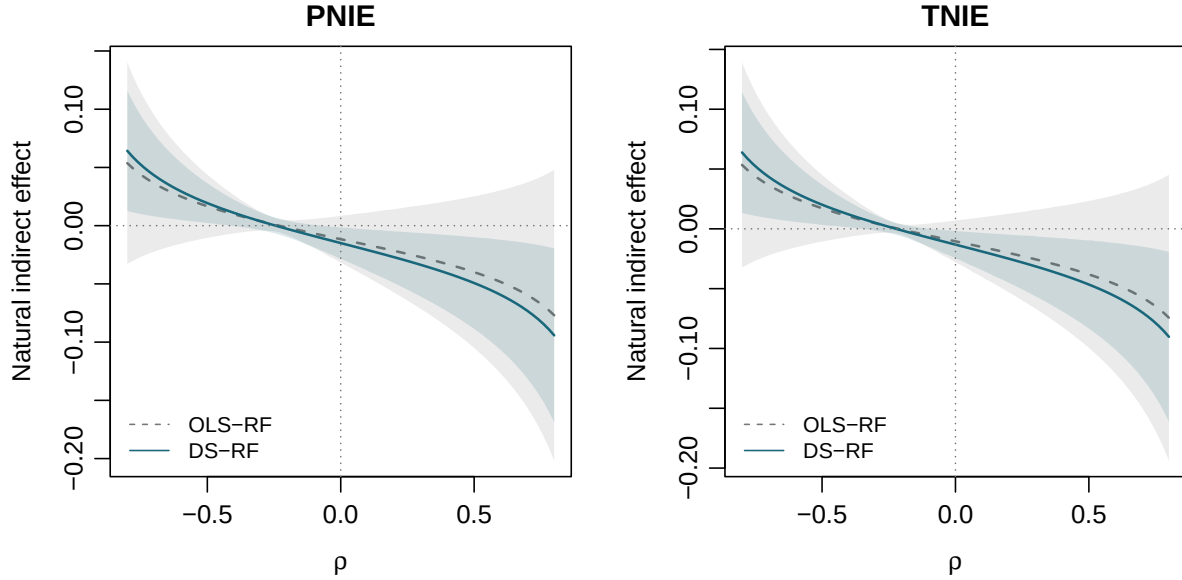


Figure S6: JOBS II reduced-form PNIE/TNIE estimates over a common analyst-specified correlation ρ . Shading gives pointwise nominal 95% intervals for each sensitivity value. The vertical reference is $\rho = 0$ and the horizontal reference is effect zero. The curves display the range of results over the specified confounding correlations.

Interval widths describe precision at a specified correlation; point-estimate zero crossings describe the confounding value needed to reverse the estimated effect. Section S8 gives their distinct influence-function representations.

S10 Reproducibility record

The source code, JOBS II data object, retained numerical results and figure-generation scripts are available at https://github.com/mijeong-kim/semi_causal_med. The repository's `README.md` describes the required software, file organization and study-specific commands; the retained seeds, fit-status records, package versions and session information document the reported analyses.

From the project root, `Rscript run_all.R` runs the supplied analysis and validation workflow and rebuilds the manuscript package when a $\text{\LaTeX}$ engine is available. To regenerate the tables, figures and PDFs from the retained results without refitting the analyses, set `JKSS_REUSE_RESULTS=1` before running the same command. The validation scripts check the effect maps and gradients, both total-effect decompositions, numerical summaries and paired-fit denominators. Separate simulation, application and document-build session records preserve the provenance of the retained results.

References

- Bickel, P. J., Klaassen, C. A. J., Ritov, Y., & Wellner, J. A. (1993). *Efficient and adaptive estimation for semiparametric models*. Johns Hopkins University Press.
- Kim, M. (2023). Appropriate use of parametric and nonparametric methods in estimating regression models with various shapes of errors. *Stat*, 12, e606. <https://doi.org/10.1002/sta4.606>
- Tsiatis, A. A. (2006). *Semiparametric theory and missing data*. Springer.
- Wang, W., Peng, X., & Tong, T. (2025). Robust and efficient mediation analysis via Huber loss. *Psychometrika*, 90(2), 734–756. <https://doi.org/10.1017/psy.2024.28>